

# What we cover...

- What is the clean core?
- What are the benefits?
- How to get to a clean core?
- Myth and Facts
- Information on where to get started

# How often do you upgrade your ERP system per year?

Show with your hands ...



**1x per year**



**2x per year**



**Let's not talk  
about it ...**

# How familiar are you with what Clean Core is?

Show with your hands ...



**Heard about it**



**Adapted  
some areas**



**Fully Clean**

# What I often year

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*"BTP is the only means of making and keeping the core clean"*

*"Fit-to-standard is the only means of making and keeping the core clean"*

*"Clean Core is only about Custom Code and Extensibility."*

*"Clean Core is only about TCO"*

# What is clean core?

It describes **modern approaches to design:**

1. business processes,
2. extensions,
3. integration scenarios, and
4. data architectures

in a

- a. stable,**
- b. upgrade-safe,**
- c. transparent** manner,
- d. along with a **separate platform to innovate** for additional differentiation



# What is clean core?

## CLEAN

Clean means **up-to-date, transparent, unmodified, consistent, efficient, and cloud compliant.**

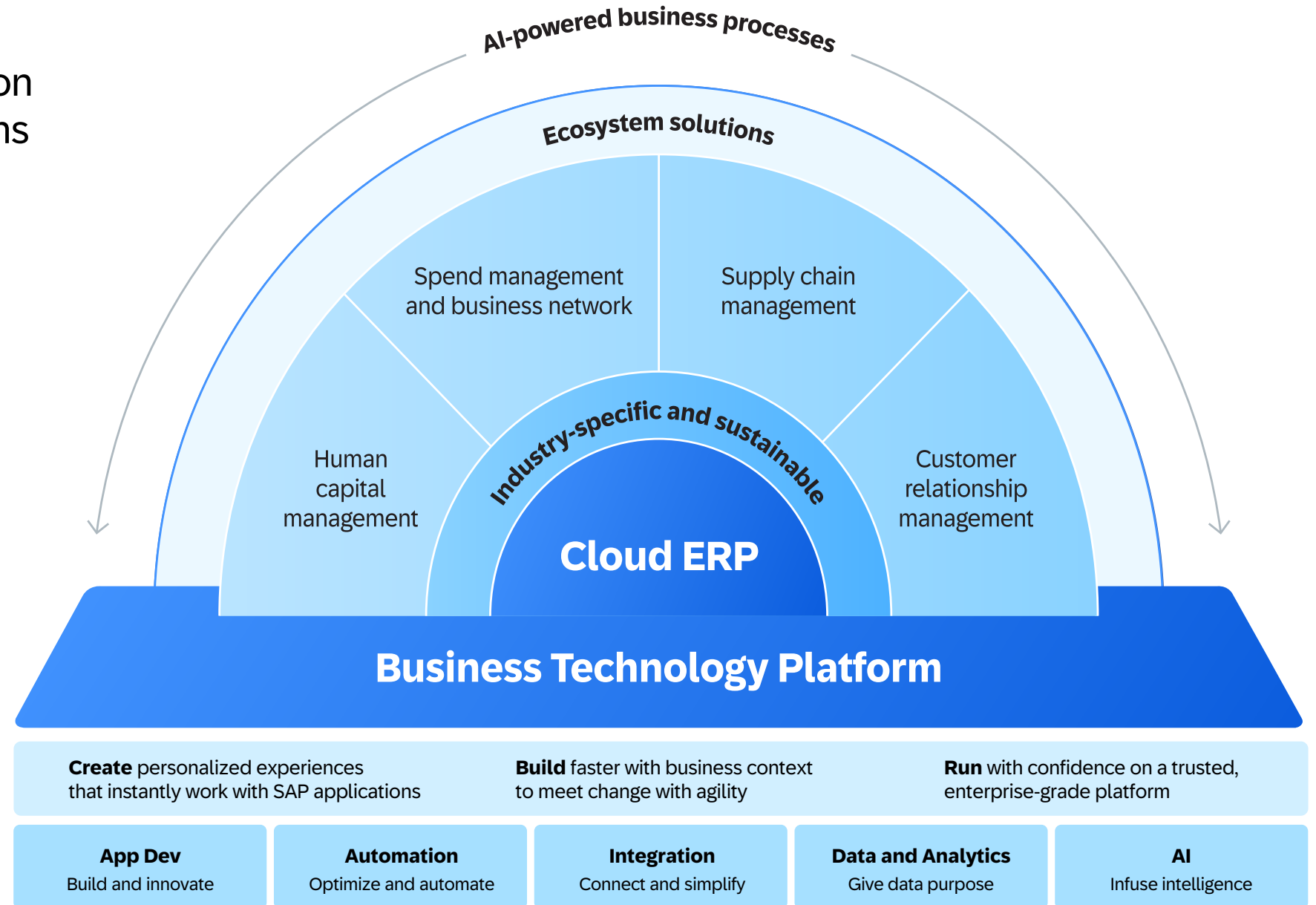


## CORE

The core describes the main aspects of an ERP system landscape, namely **extensibility, processes, data, integration, and operation.** Capabilities of the core depend on the implemented software stack.

# SAP BTP

Unleash digital innovation  
across business functions





**“A new public cloud system is clean  
per the definition.”**

PUBLIC

# Benefits of clean core

**1. Agility and flexibility**

**2. The ability to leverage the latest innovations (stay current)**

**3. Efficient and more secure operations**

**4. Data to value**

# How to clean core

Following a clean core strategy means:

- **Reducing** data and process **inconsistencies**
- Establishing suitable **housekeeping processes** and **governance structures**
- Maximizing the use of **standard functionality** and its ROI
- Building **cloud-compliant** extensions that are integrated using **stable interfaces**
- **Minimizing** customization **complexity**



# Keep the **extensions** clean

## Main Aspects:

- ✿ Extensions should be **avoided** when possible
- ✿ Create **cloud compliant Extensions**,
- ✿ **Custom extensions** do not break an upgrade and **upgrades** do not break an extension
- ✿ Leverage 1: **on the stack** & 2. side-by-side with **SAP BTP**

Keep the **extensions** clean

1

Extensibility

Keep the **data** clean

2

Data

Keep the **Integrations** clean

3

Integrations

Keep the **processes** clean

4

Processes

Keep the **operations** effective and efficient

5

Operations

# What makes extensions clean core compliant?

## What is a “good” extension?

### Main aspects

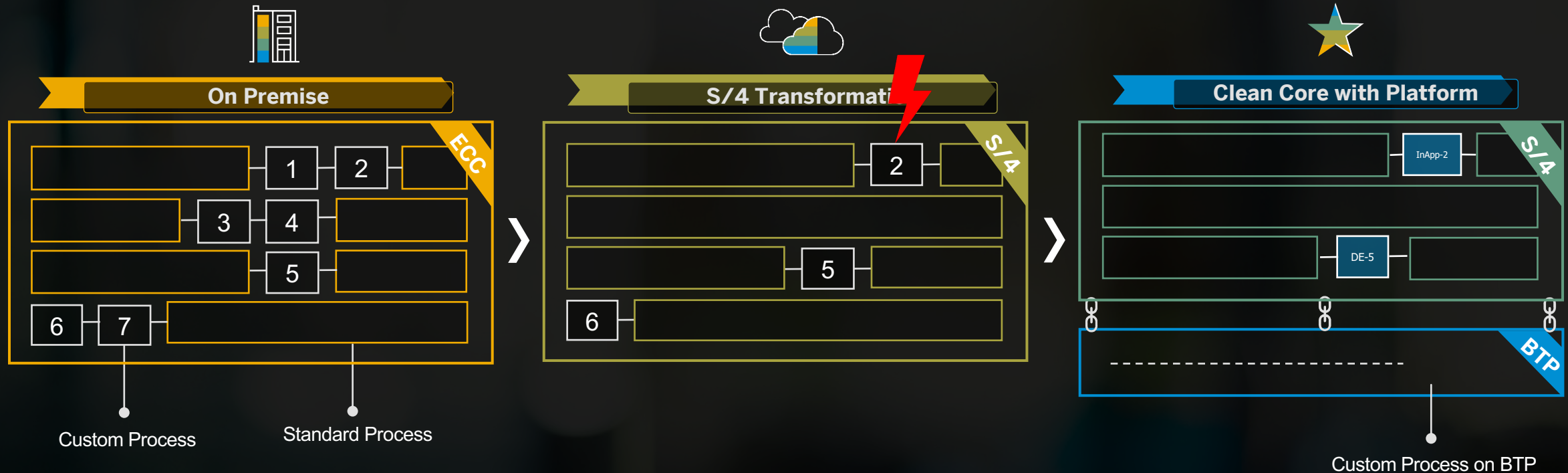
- **Avoid** extensions when possible
- Set up a **strong governance** to create decoupled extensions in a way that they would work in the cloud (three-tier model)
- Separate extensions by leveraging released APIs – **custom extensions** do not break an upgrade and **upgrades** do not break an extension\*
- Leverage the full capabilities of extensibility **on the stack** as well as side by side **with SAP BTP**
- Create **technical debts** only as informed decision

\*Ensuring upgrade stability can be a short-term workaround for transforming a whole application from traditionally developed code into cloud-compliant (Tier 1) extensions.

### How to achieve

- Establish a governance model – a clearly defined process with high demands to approve any extension.
  - Prefer standard over custom development by leveraging fit-to-standard best practices.
  - Avoid custom code where possible. Don’t extend for rarely needed use cases.
  - Prefer “clean” extension options over “unclean” ones.
  - Use [SAP Application Extension Methodology](#) and [extensibility guidance](#) to identify the best path in your landscape.
- If you need to extend, a clear separation is key.
  - Only access standard objects through [released](#) and stable APIs (either remote or locally; access for reading and changing access possible).
  - Choose only “clean” tools or environments and extensibility options (in SAP S/4HANA: key user, developer, or side-by-side extensions).
- Choose extension domain based on requirements only.
  - SAP BTP automatically decouples extension but is not the only “clean” approach.
  - Do not extend in the core simply because “we always do so.”
- Enable awareness.
  - In on-premise installations, you can actively decide to develop some extensions not clean core, as long as they are documented and informed decisions (use cases: copy routines; API not available, and more).
  - Mitigate missing APIs in private cloud or on premise by using wrappers as described in [ABAP Cloud API Enablement Guidelines for SAP S/4HANA Cloud, private edition, and SAP S/4HANA](#).
  - Create requests for APIs using a customer influence tool (for [public](#) or [private](#) cloud editions).

# App Dev – Why and how to achieve a Clean Core?



## Extensibility Options in S/4HANA

- InApp = In-App Extensibility
- DE = Developer Extensibility

## Extensibility Options

1

**In-App Extensibility  
S/4HANA**

2

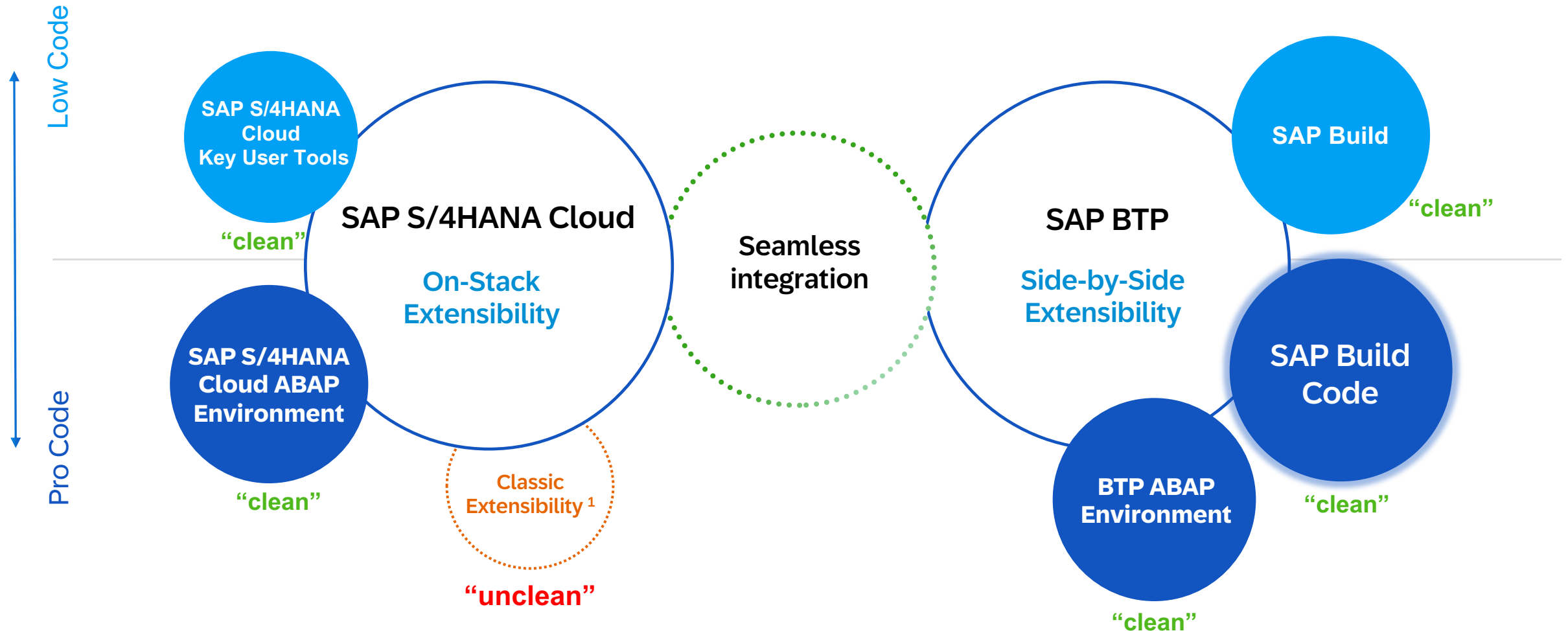
**Developer Extensibility  
S/4HANA**

3

**Side-by-Side Extensibility  
SAP BTP**

# Clean Core Extensibility Options for SAP S/4HANA

Core Solution Extensions ← → Side-by-Side Extensions





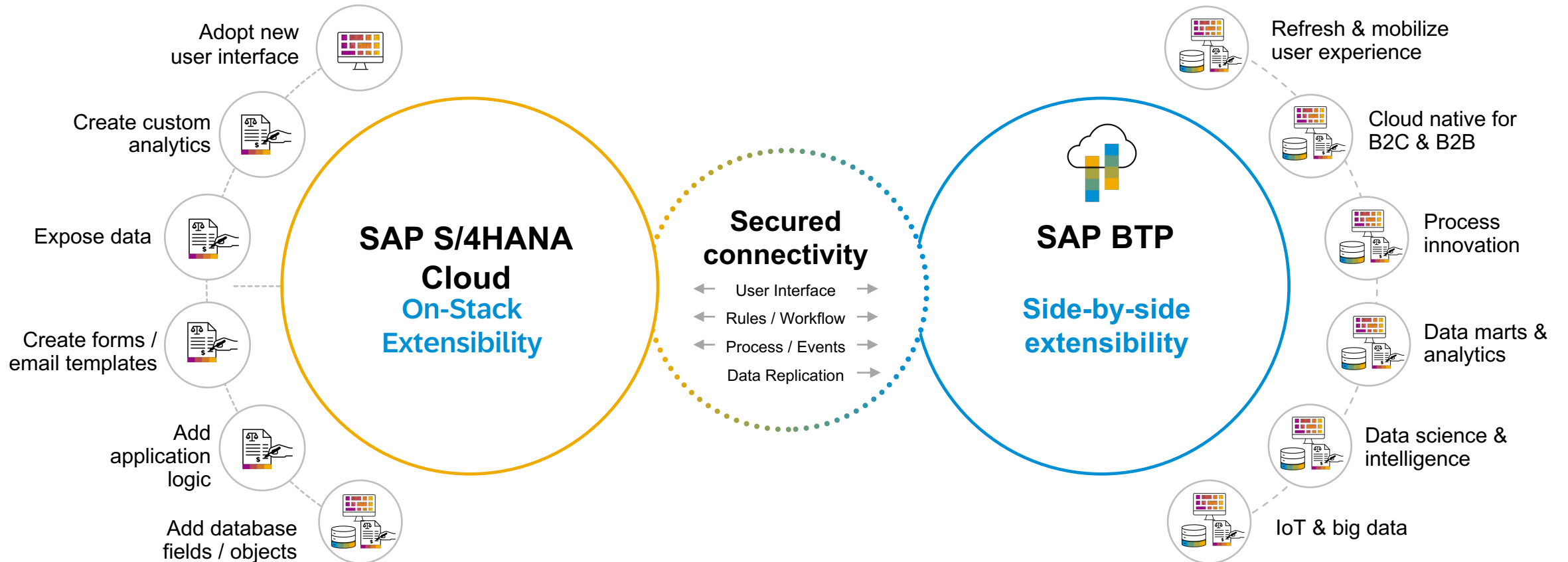
**When to use use which tool?**

# Extensibility Options Today

In-App Extensions



Cloud Extensions



# When to use use which tool?

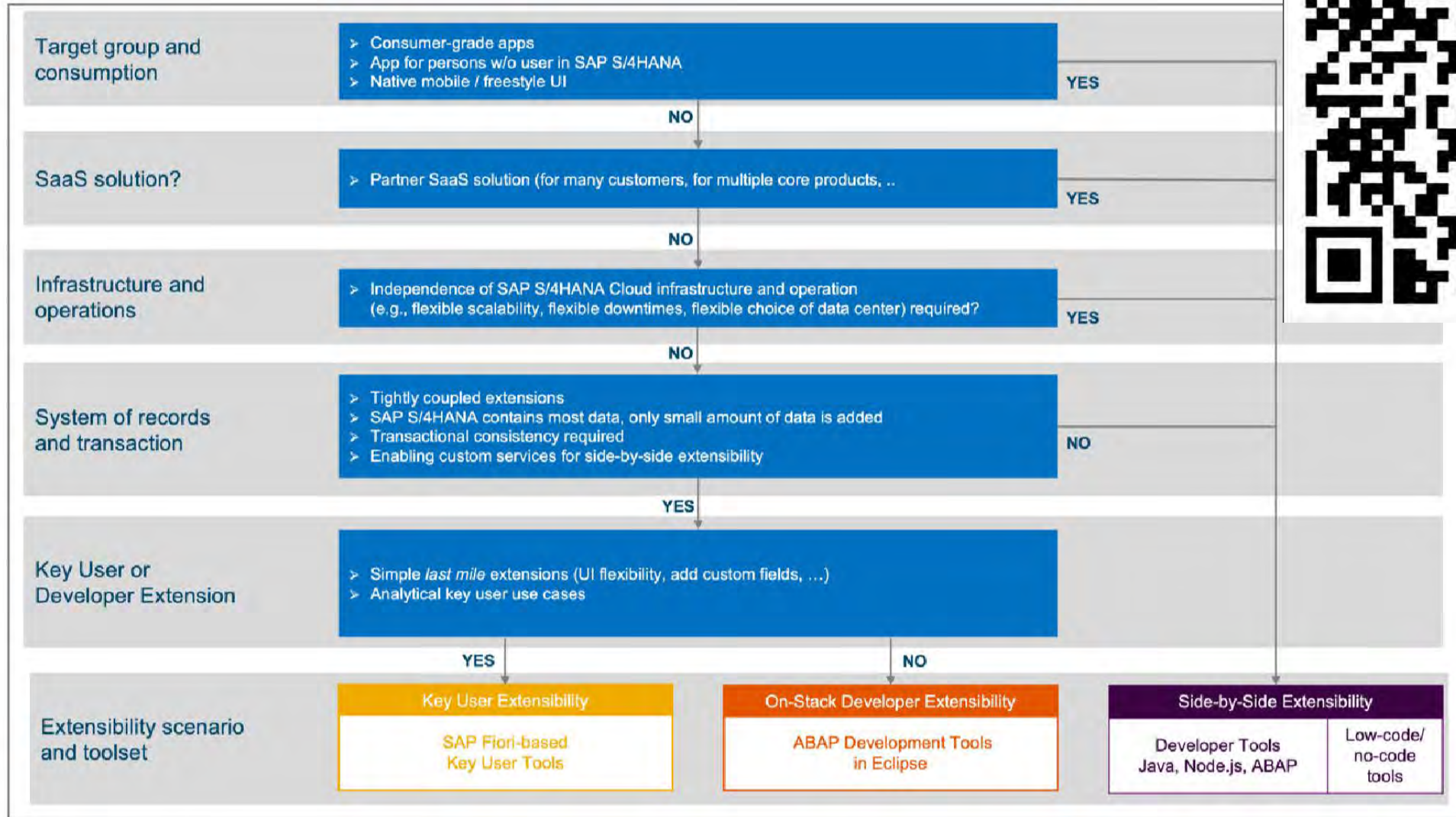
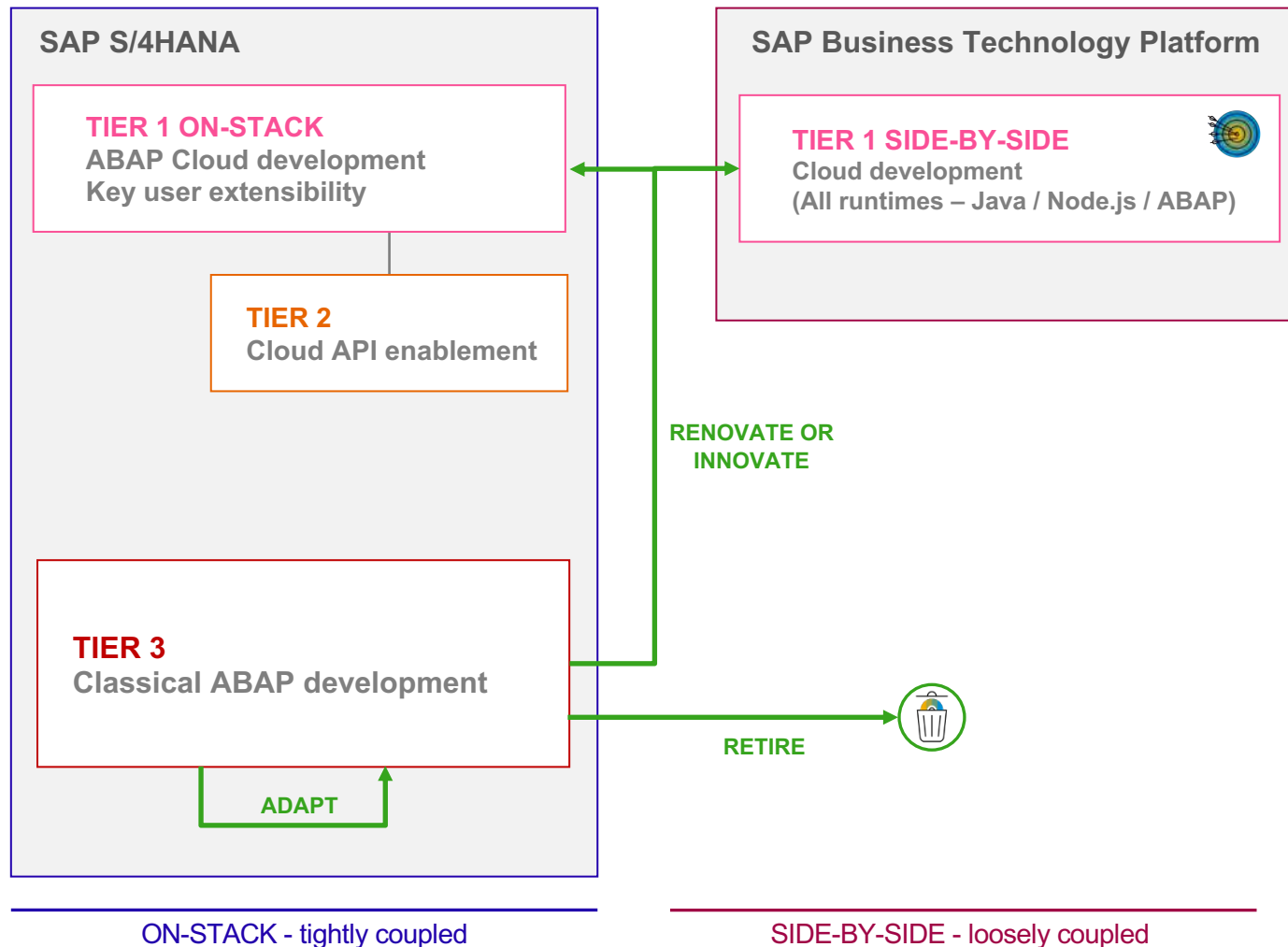


Figure 3.1 - Sequence diagram on how to find the right extensibility options.

**What if no API is available?**

# 3-tier extensibility model for SAP S/4HANA private cloud and on-premise



## TIER 1 – Cloud extensibility model

Development of cloud-ready and upgrade-stable applications and extensions

**Same development model as used in SAP S/4HANA Cloud, public edition**

Default for new extensions and custom apps

## TIER 2 – Cloud API enablement

Extends and enables tier 1 for private cloud and on-premise  
Mitigates missing public SAP APIs or extension points:

- Develop custom wrapper objects for not released SAP objects to be used in tier 1
- Retire wrapper once a released SAP API is available

## TIER 3 – Classic ABAP extensions

Legacy/existing custom ABAP code or new on-stack extensions code that cannot follow the rules of tier 1 and 2

Avoid and reduce the content in tier 3

# Local APIs – Current Status and Future Direction



- Success of all **Extension Projects** depends on a rich set of local APIs

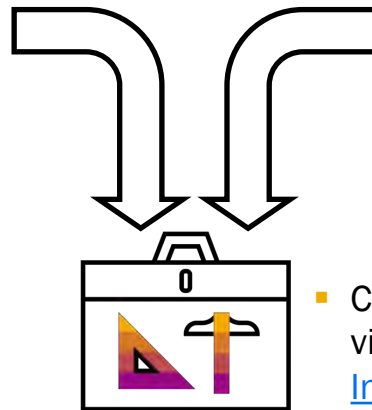
## Available APIs



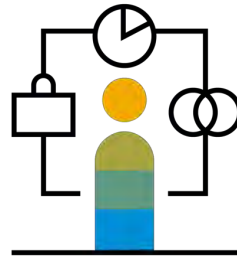
## Upcoming New APIs



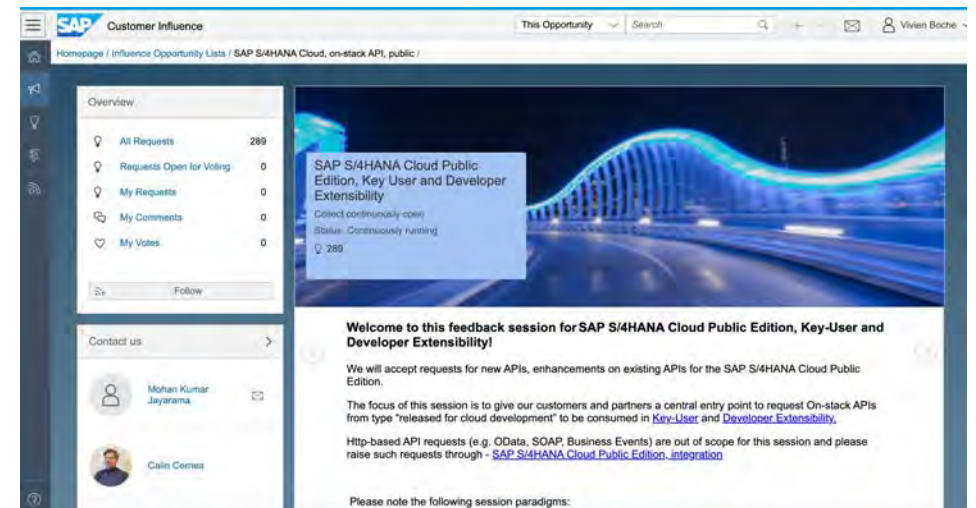
- SAP continuously delivers new local APIs with high priority



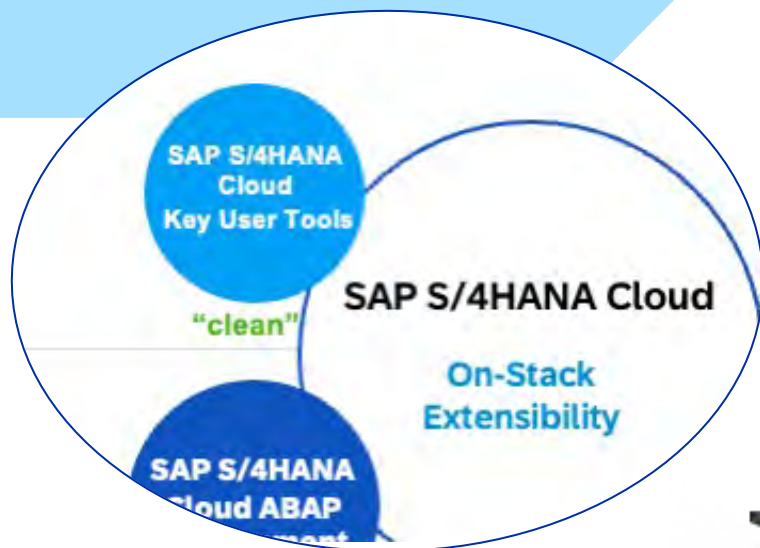
Additional  
Local APIs



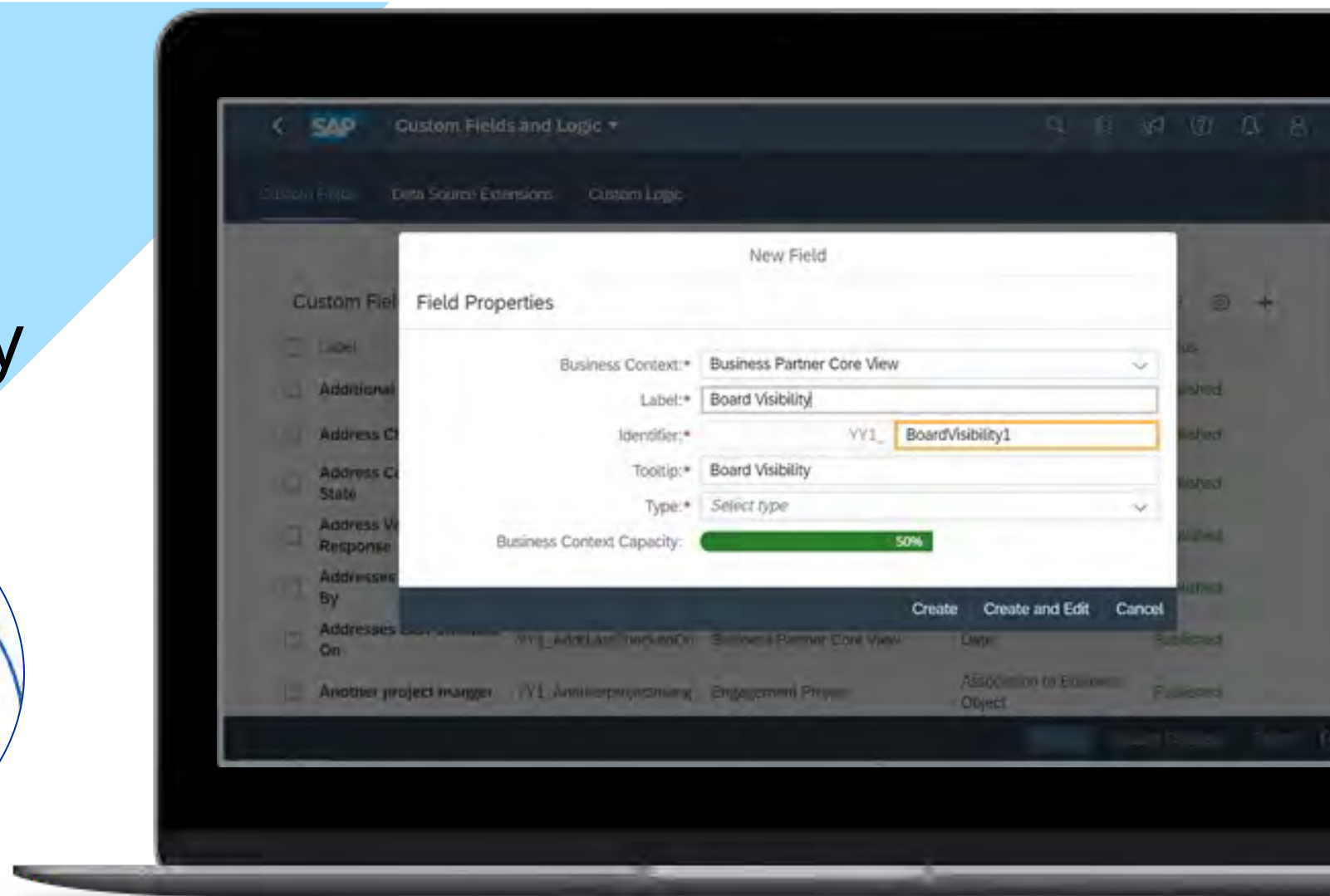
- Customers can request APIs via dedicated [Customer Influence Session](#)
- Shipment of API via Continuous Feature Delivery (CFD) or with next Release



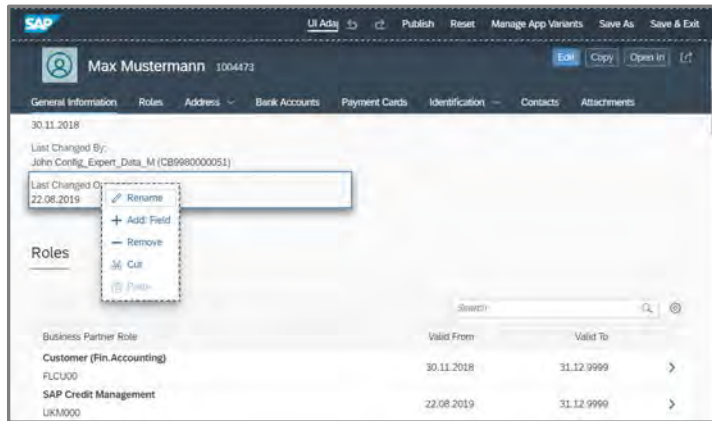
# Key User Extensibility



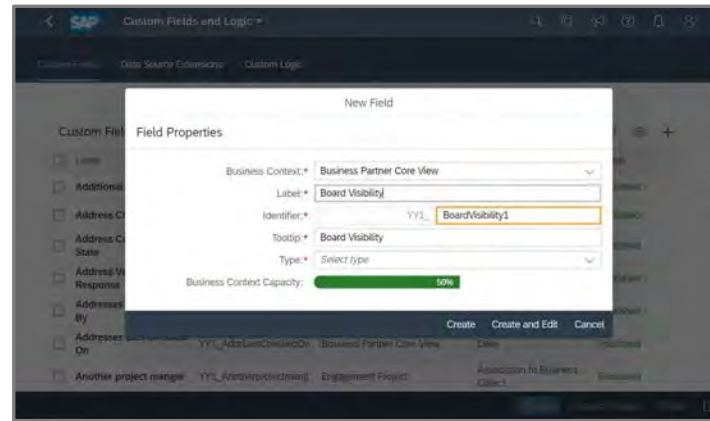
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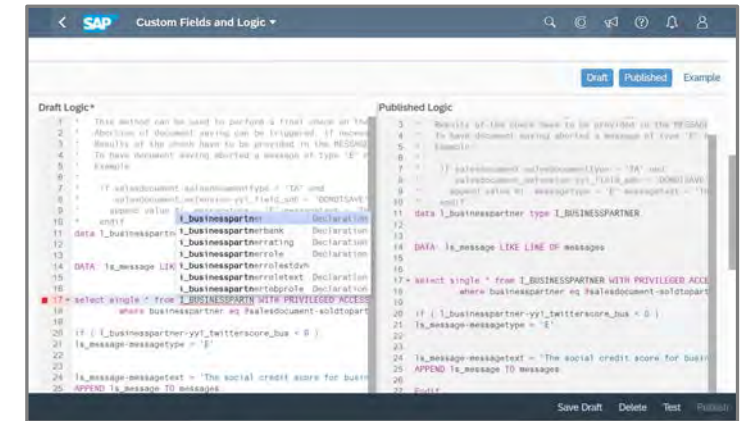




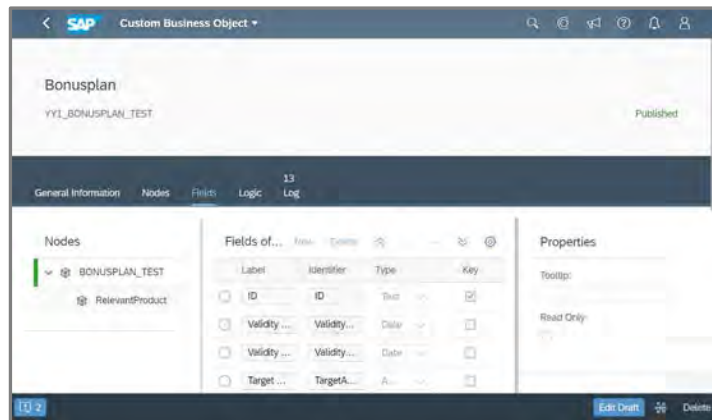
UI Adaptation Mode – Adaptation mode with drag & drop configuration



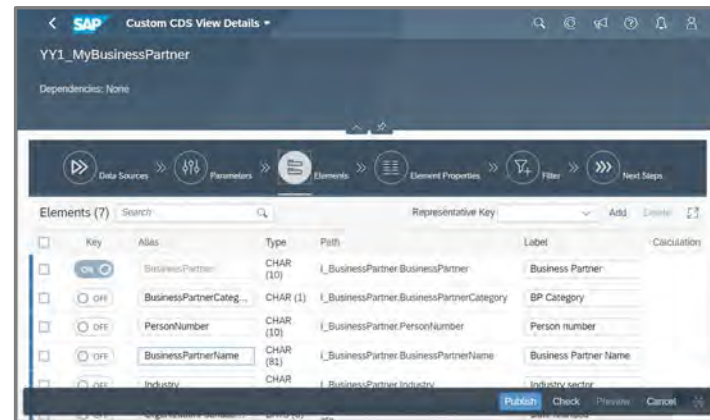
Custom Fields SAP Fiori App



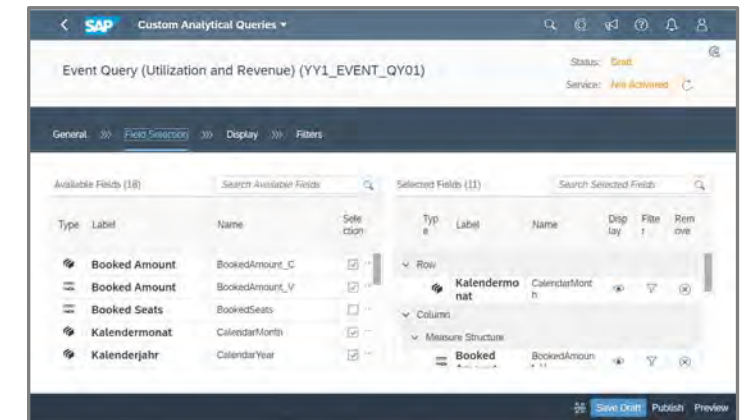
Custom Business Logic SAP Fiori App



Custom Business Objects SAP Fiori App



Custom CDS Views SAP Fiori App

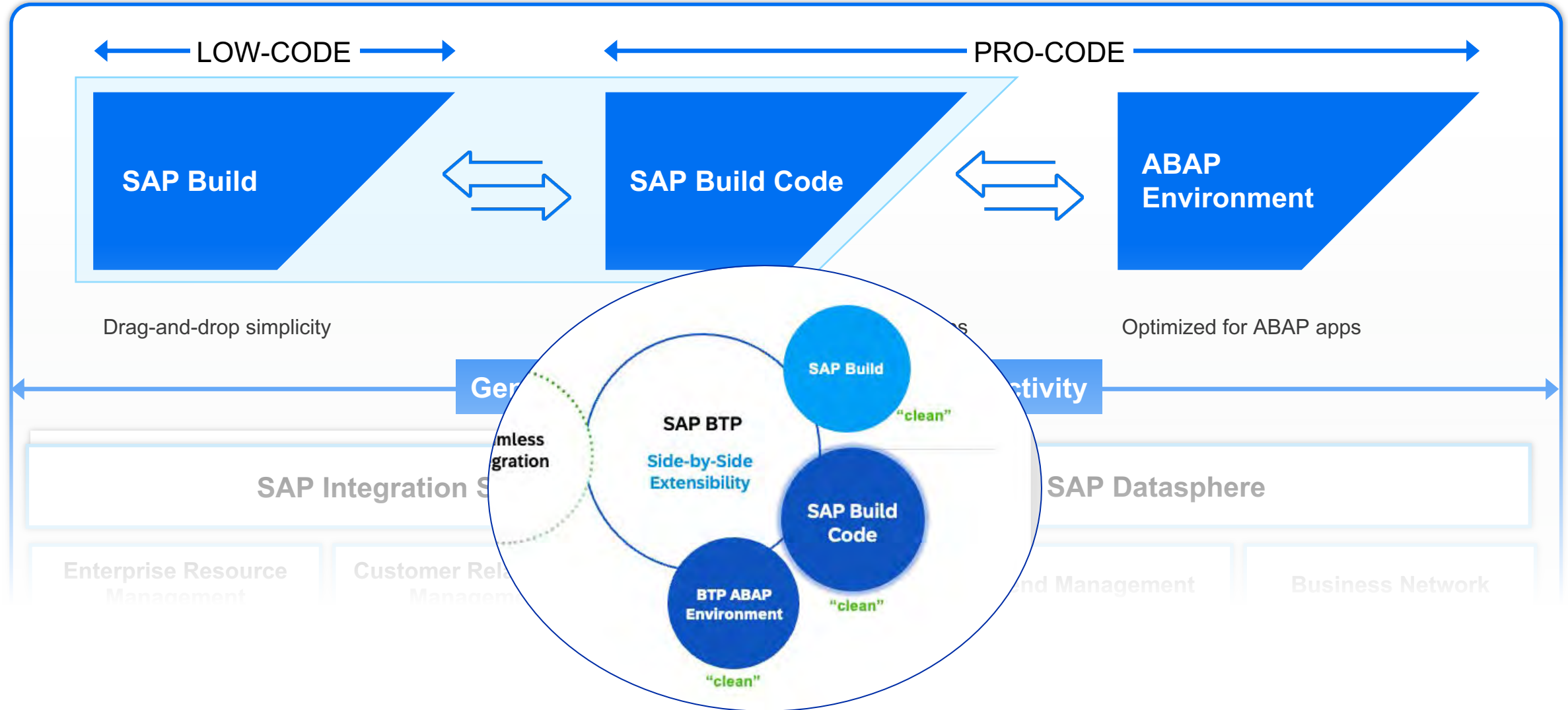


Custom Analytical Queries SAP Fiori App



# Application development and automation on SAP BTP

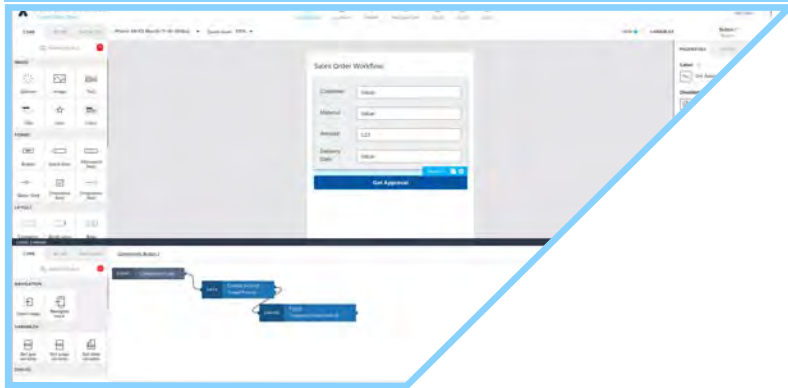
Integrated solutions enabled by generative AI



# SAP Build

SAP Build enables everyone no matter their skill level to create and augment enterprise applications, process automations and business sites with drag-and-drop simplicity.

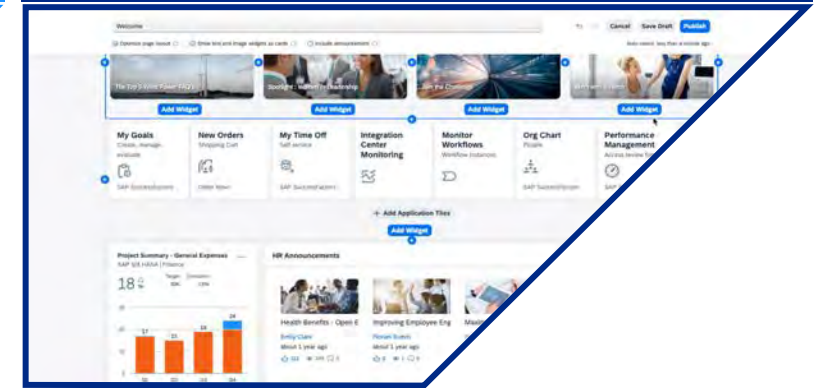
## SAP Build Apps



## SAP Build Process Automation



## SAP Build Work Zone



Build visually

Integrate seamlessly

Collaborate in fusion teams

# SAP Build Apps

Build full-stack enterprise-apps in minutes - absolutely zero coding required

## Drag-and-drop the user interface

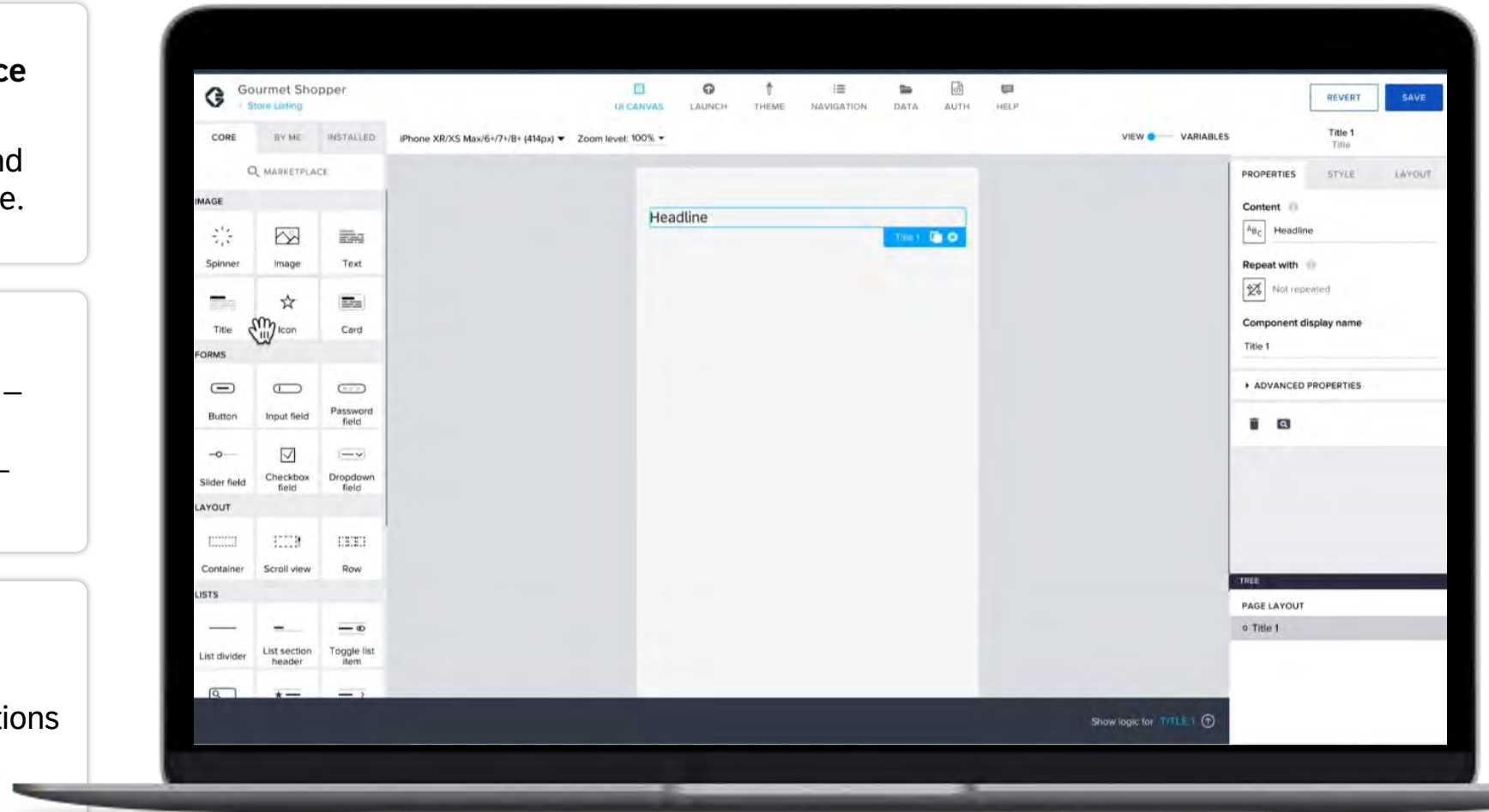
Access a pallet of user interface controls, drag them into place, and preview the final result in real time.

## Create any logic without code

Drag-and-drop any logic function – sensors, file operations, data functions, animations, and more – onto a canvas.

## Easily integrate data

Easily add your own data integrations or get started with some of ours.



# Pif Paf Alimentos

A mobile application  
for maintenance  
technicians using  
SAP Build Apps

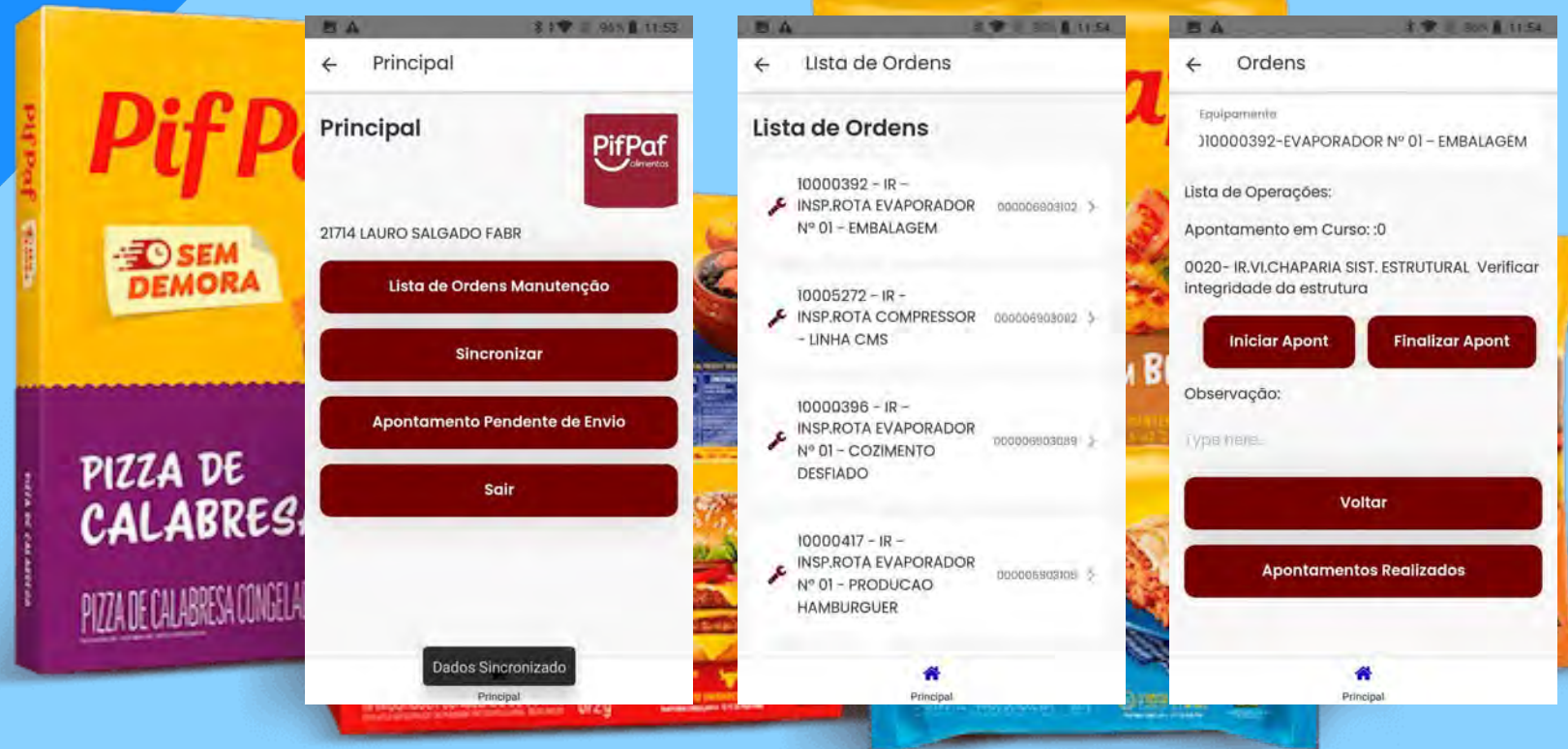


 SAP  
Innovation  
Awards 2024



# Pif Paf Alimentos

A mobile application for maintenance technicians using SAP Build Apps



3 weeks

To go-live with a mobile application

30%

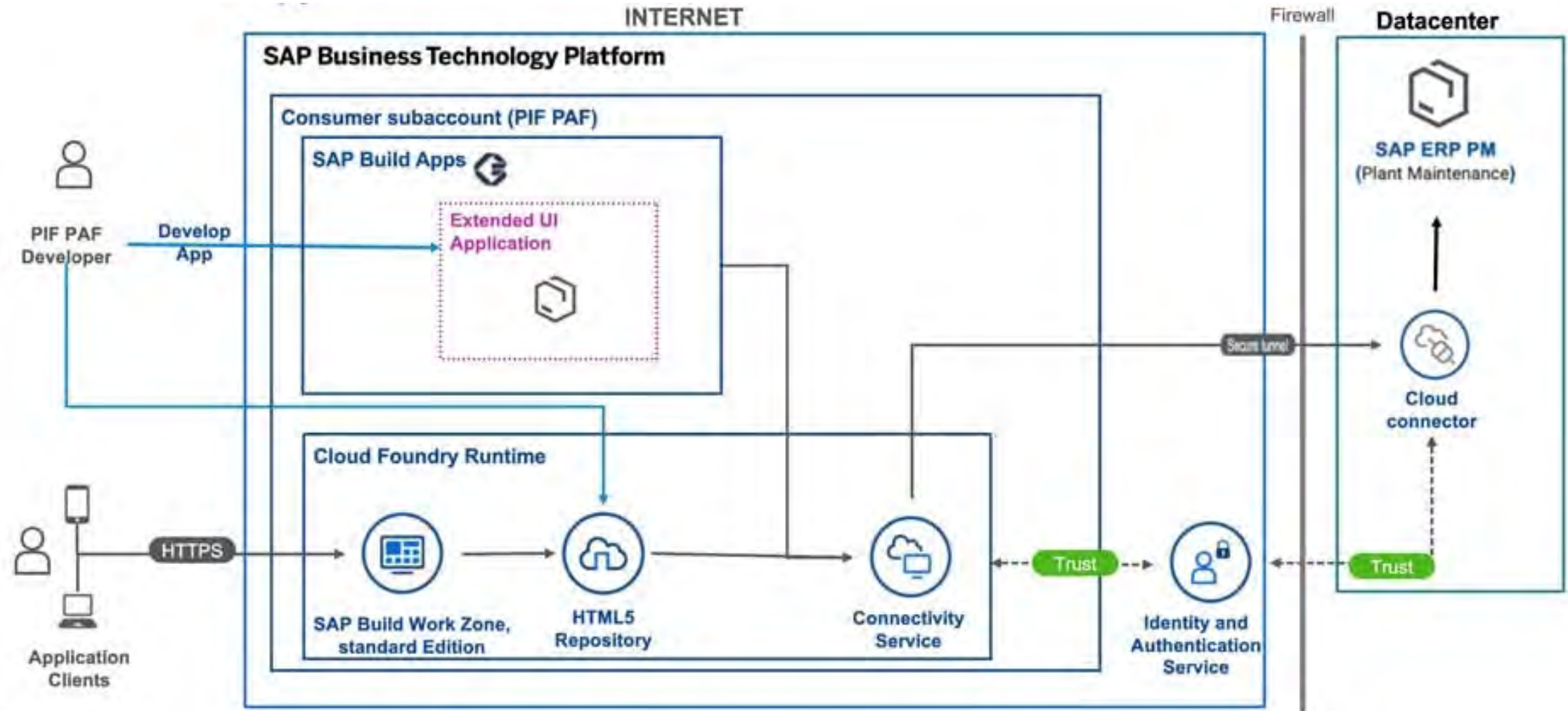
Reduction in time and cost of technician data entry

10%

Reduction in paper waste

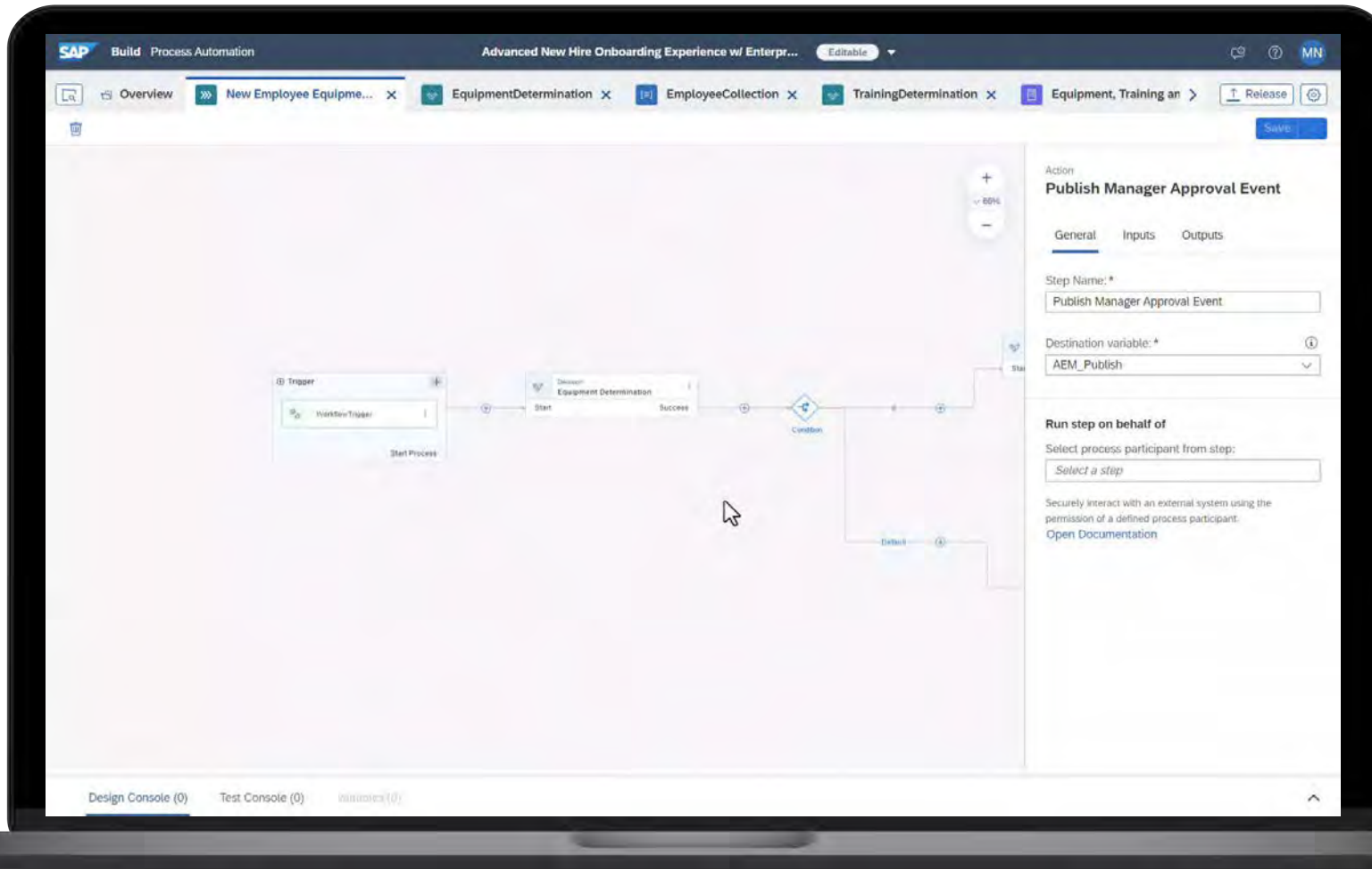
# Pif Paf Alimentos

## Architecture details



# Automate visually

Simplify process and task automation



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**Automate processes and tasks with drag-and-drop simplicity**

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**Workflow management, robotic process automation (RPA), and embedded AI in a single solution**

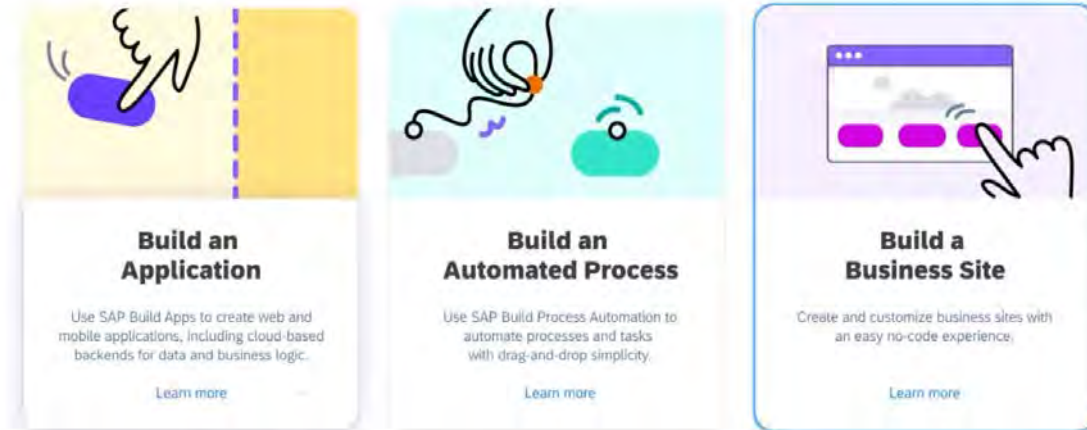
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**Leverage process expertise without coding**

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# Build simpler and faster – Empower everyone to build and personalize business sites visually

What would you like to do?



Build branded sites, pages and workspaces without writing any code

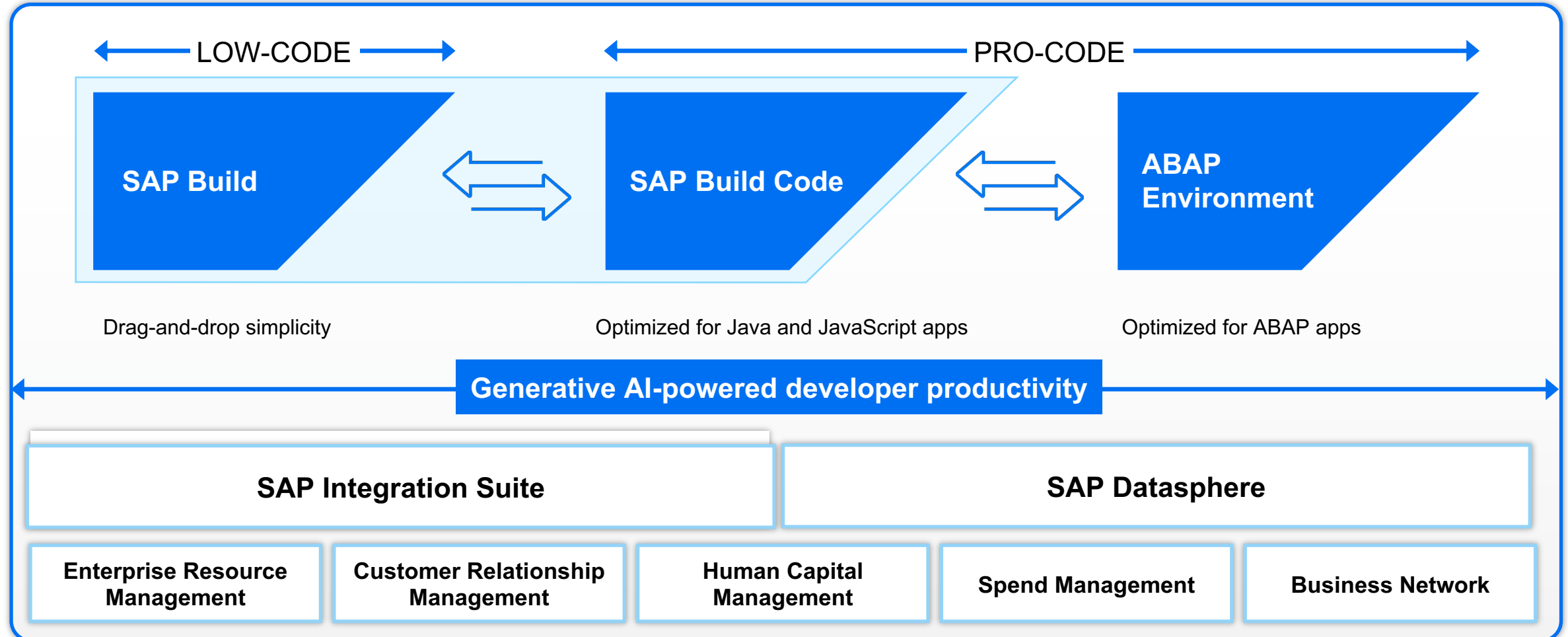
Personalize sites and target content to users based on profile information (location, department, role)

Quickly add new users and deploy business site updates



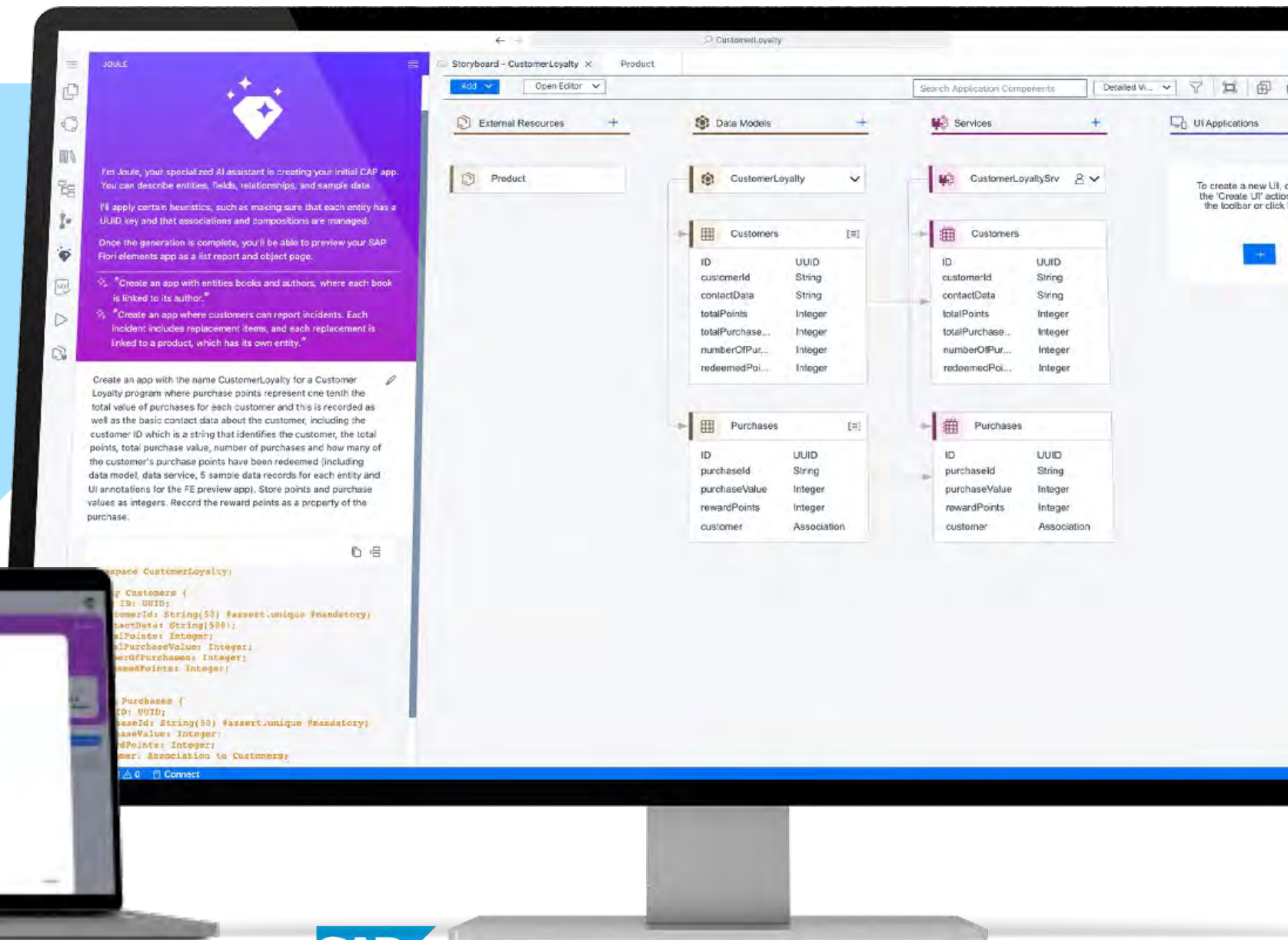
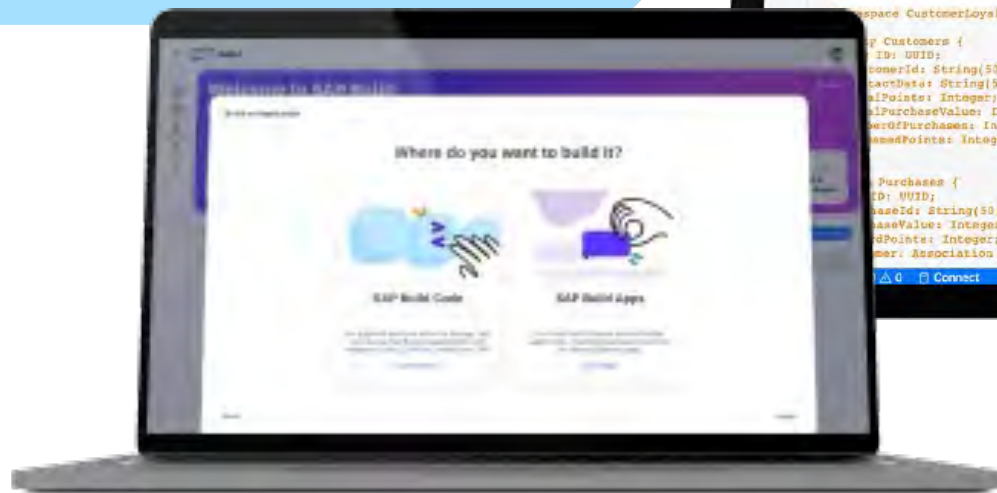
# Application development and automation on SAP BTP

Integrated solutions enabled by generative AI



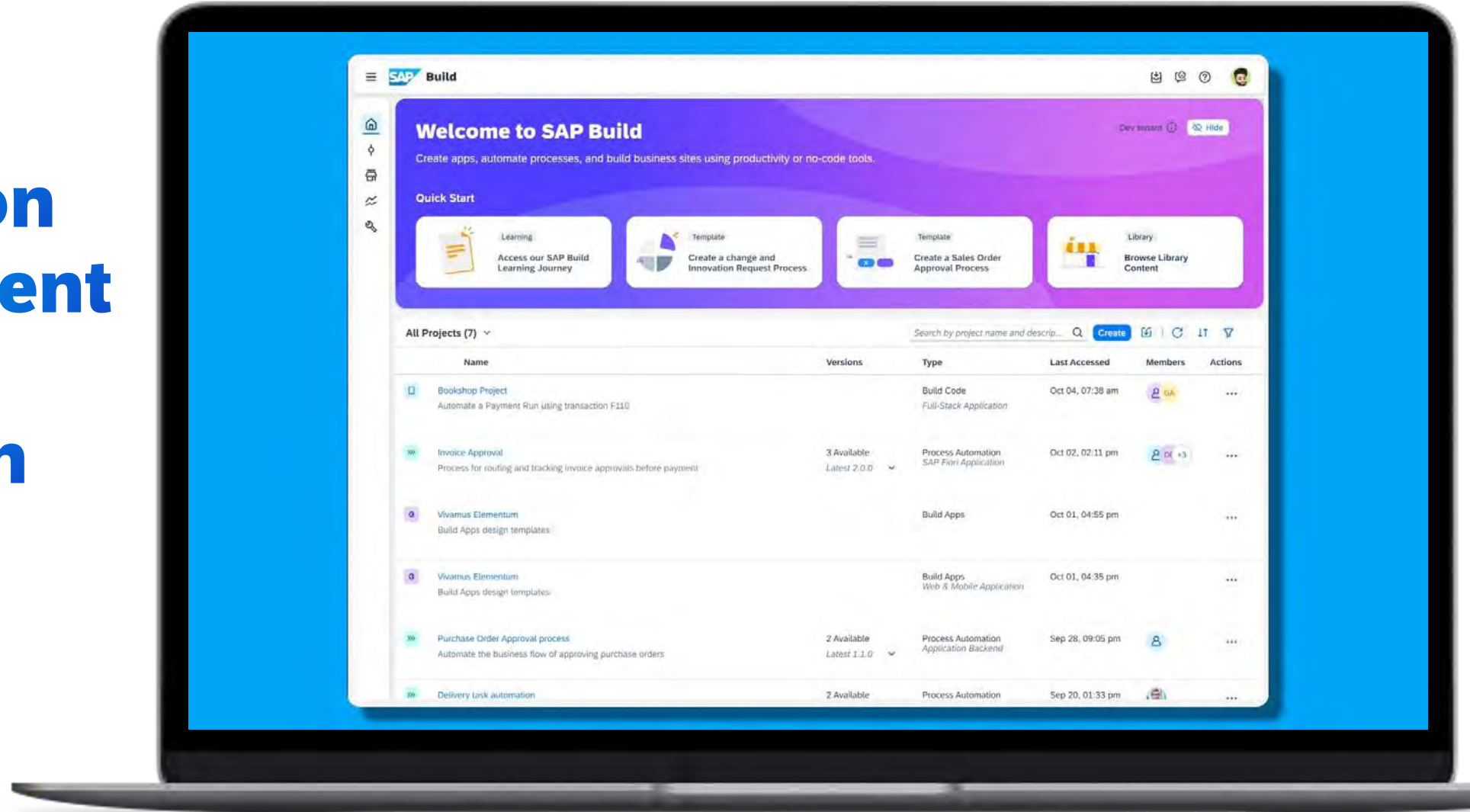
# SAP BUILD CODE

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# SAP Build Code

**Faster  
Application  
development  
via code  
generation**



# SAP Build Code

Application development for all key extension use cases in SAP environments



Extend SAP Solutions



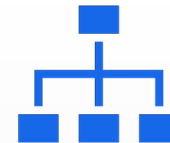
Build Fiori web apps with  
SAPUI5 and SAP Fiori elements



Build and run  
mobile apps



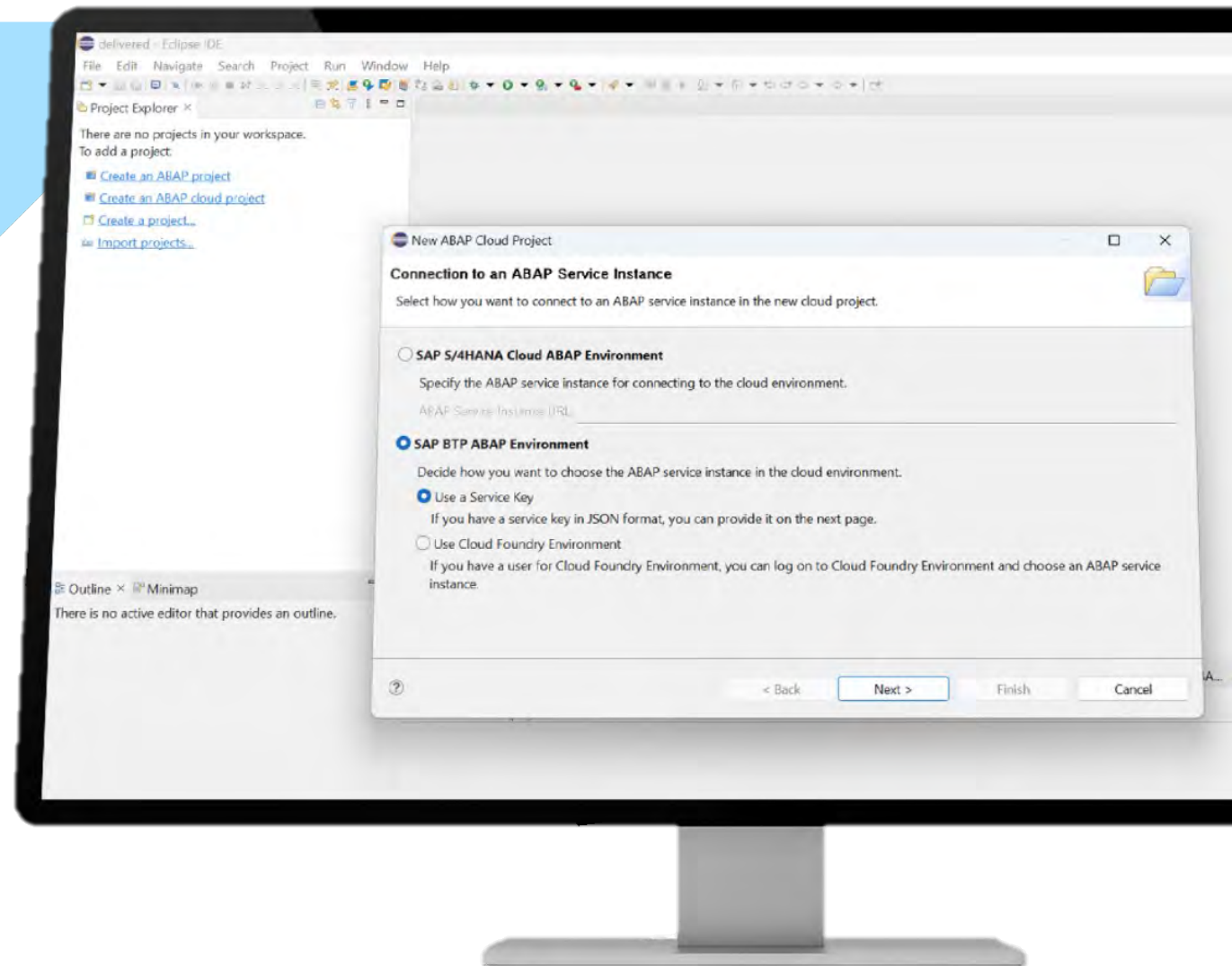
Develop SAP HANA  
Native applications



Build multi-tenant  
SaaS applications

# SAP BTP, ABAP Environment

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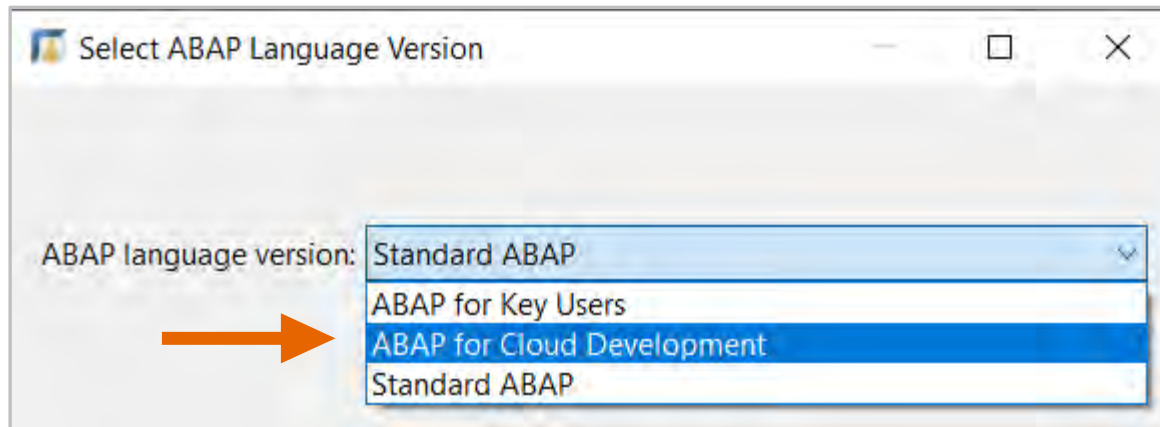


# SAP S/4HANA, private cloud and on-premise editions

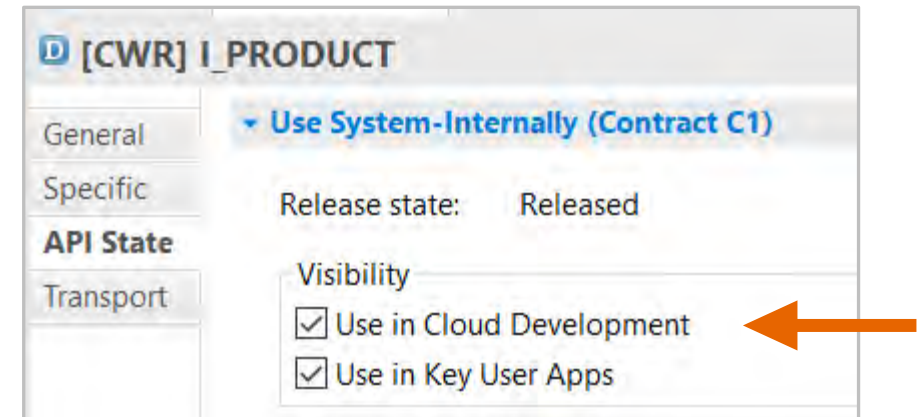
## ABAP language version and public SAP APIs

### ABAP cloud development in the private cloud and on-premise editions of SAP S/4HANA

- Switch ABAP language version for development object or package
- Inspect the “Release state” for used APIs and objects



Switch from classic ABAP extensibility (standard ABAP) to ABAP cloud (ABAP for cloud development)



SAP released the CDS view for *ABAP cloud development*



# ABAP Cloud – IDE and API examples

## Custom ABAP on SAP S/4HANA Cloud

ABAP development tools in Eclipse  
Cloud-optimized ABAP language  
Proven ABAP transport management

## Access to public SAP APIs ONLY – otherwise, syntax error!

No access to old Dynpro APIs  
No direct select on the MARA table from SAP

## Local APIs from SAP S/4HANA

SELECT products from  
SAP S/4HANA tables using the public  
I\_Product CDS view

The screenshot displays the SAP S/4HANA Cloud ABAP IDE environment. At the top right, a diagram shows the architecture: SAP S/4HANA (CLOUD) connects to public interfaces, which then connect to the SAP S/4HANA CLOUD ABAP ENVIRONMENT / ABAP PLATFORM. A callout box indicates 'ABAP CLOUD DEVELOPMENT ON SAP S/4HANA'.

The main editor shows the following ABAP code:

```
1 CLASS zcl_bgtest DEFINITION
2   PUBLIC
3   FINAL
4   CREATE PUBLIC.
5
6   PUBLIC SECTION.
7     INTERFACES if_oo_adt_classrun.
8   PROTECTED SECTION.
9   PRIVATE SECTION.
10  ENDCLASS.
11
12
13 CLASS zcl_bgtest IMPLEMENTATION.
14
15   METHOD if_oo_adt_classrun~main.
16
17     SELECT CountryName FROM I_CountryText WHERE Language = @sy-langu ORDER BY CountryName INTO @DATA(l_countryname).
18     out->write( l_countryname ).
19     ENDSELECT.
20
21     CALL FUNCTION 'POPUP_TO_CONFIRM'.
22     SELECT matnr FROM mara INTO TABLE @DATA(l_matnr).
23
24     SELECT ProductExternalID FROM I_Product INTO TABLE @DATA(l_matnr_new).
25
26   ENDMETHOD.
27
28 ENDCLASS.
```

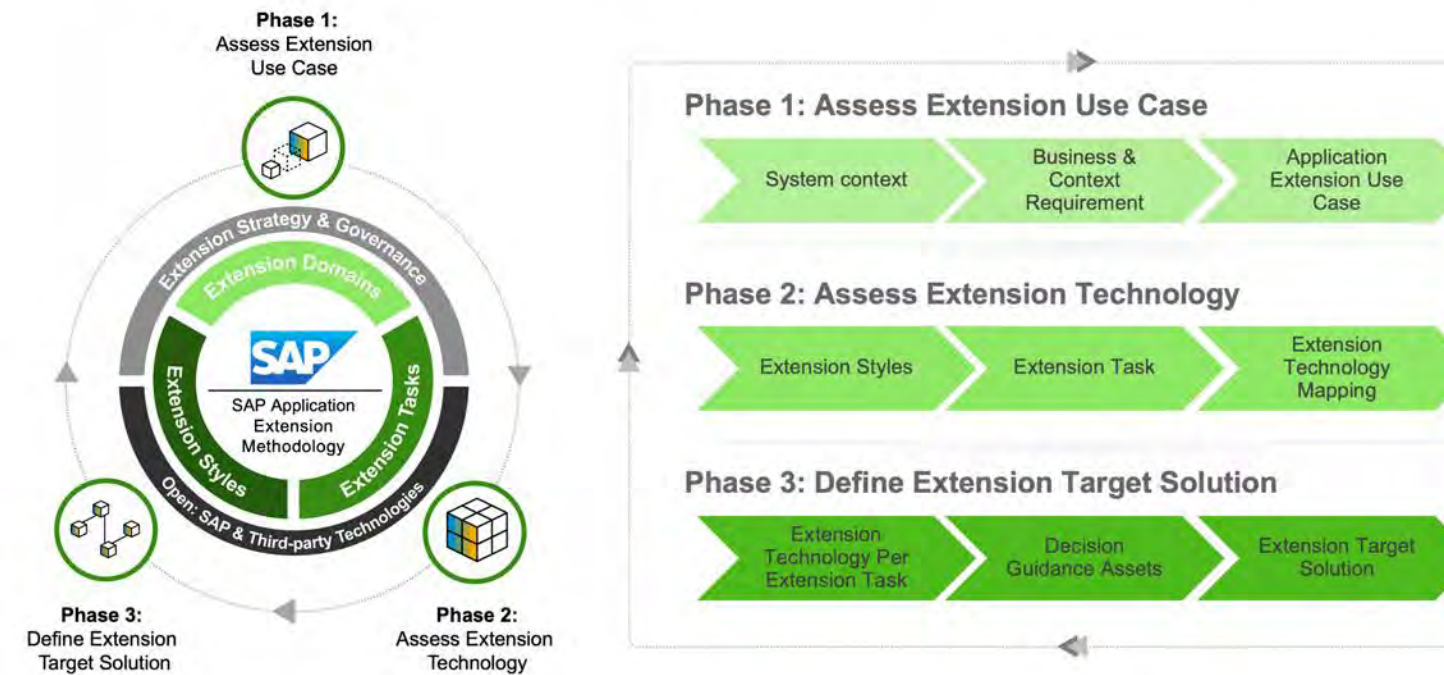
The bottom of the screenshot shows the Eclipse IDE interface with the 'Problems' tab active, displaying three errors:

| Description                                                                     | Resource          | Path                    | Location | Type             |
|---------------------------------------------------------------------------------|-------------------|-------------------------|----------|------------------|
| ✖ The use of Function Module POPUP_TO_CONFIRM is not permitted.                 | zcl_bgtest.aclass | /CWR_EN/.adt/classli... | line 21  | ABAP Syntax C... |
| ✖ Use of Table MARA is not permitted. See object documentation for replacement. | zcl_bgtest.aclass | /CWR_EN/.adt/classli... | line 22  | ABAP Syntax C... |



# SAP Application Extension Methodology

## SAP Application Extension Methodology – Overview: Phase 1 - 3



## [SAP Application Extension Methodology](#)



The [SAP Application Extension Methodology](#) helps you to define, document and execute an enterprise extension strategy for your organization.



# Keep the **data** clean

## Main Aspects:

- ✿ Modern data requirements
- ✿ Data volume
- ✿ Outdated, unused or redundant information
- ✿ Personal master data

Keep the **extensions** clean

1

Extensibility

Keep the **data** clean

2

Data

Keep the **Integrations** clean

3

Integrations

Keep the **processes** clean

4

Processes

Keep the **operations** effective and efficient

5

Operations

# What makes data clean core compliant?

What is “clean” data?

## Main aspects

### Data quality

(Configuration, master, and transactional data)\*

- Accuracy
- Timeliness
- Completeness
- Validity
- Consistency
- Uniqueness

### Data volume efficiency

(Master and transactional data)

- Optimized memory and disk consumption
- No outdated, unused, or redundant information
- Data lifecycle management (creation, updates, end of life)

### Data privacy compliance

- Storing and processing personal master data only with justifiable purposes

\***Configuration data:** General data that defines the organization's structure and is of static nature (such as company code, plants, purchase organisations, controlling area, or sales area); **Master data:** Consistent and uniform set of identifiers and extended attributes that describe the core entities of the enterprise, such as customers, vendors, products and general ledger accounts; **Transactional data:** Information directly derived as a result of transactions, this data always has a time dimension, a numerical value, and refers to one or more (master data) objects

## How to achieve

### For data quality:

- Analyze and define data quality measures for critical data objects. SAP provides data quality measures for several standard data objects.
  - If necessary, involve SAP or third-party vendors (such as CDQ) for getting further help and advice.
- Establish a “get clean” process.
  - Define a tool-based, reusable data cleansing process (such as through the [SAP Master Data Governance](#) application [consolidation], [SAP Information Steward](#) software, [SAP Data Intelligence](#) solution, [quality services for SAP BTP](#), or other third-party tools) for deduplication, generation of best records, and more.
- Establish a continuous “keep clean” process.
  - Define a validation rule framework, approval process, and automated distribution framework to connected receiver systems for newly created, changed, or deleted data records (such as using [SAP Master Data Governance](#) or [SAP Master Data Integration service](#)).
  - Define continuous data monitoring.
- Adhere to [SAP One Domain Model](#) (universal language across SAP systems) and [SAP Data and Analytics Advisory Methodology](#).

### For data volume efficiency (from creation until end of life):

- Enable efficient continuous analysis and monitoring of the database by reducing outdated, unused, or redundant data.
- Define archiving or deletion of data-tiering processes to improve efficiency of the database (for example, using the SAP Information Lifecycle Management component).

### For data privacy compliance:

- Analyze data usage to clarify business purposes of collecting and processing personal master data.
- Establish policies to govern personal master data lifecycle (using SAP Information Lifecycle Management).

## Benefits

**Reliable results** when using data in processes and analytical applications (data to value)

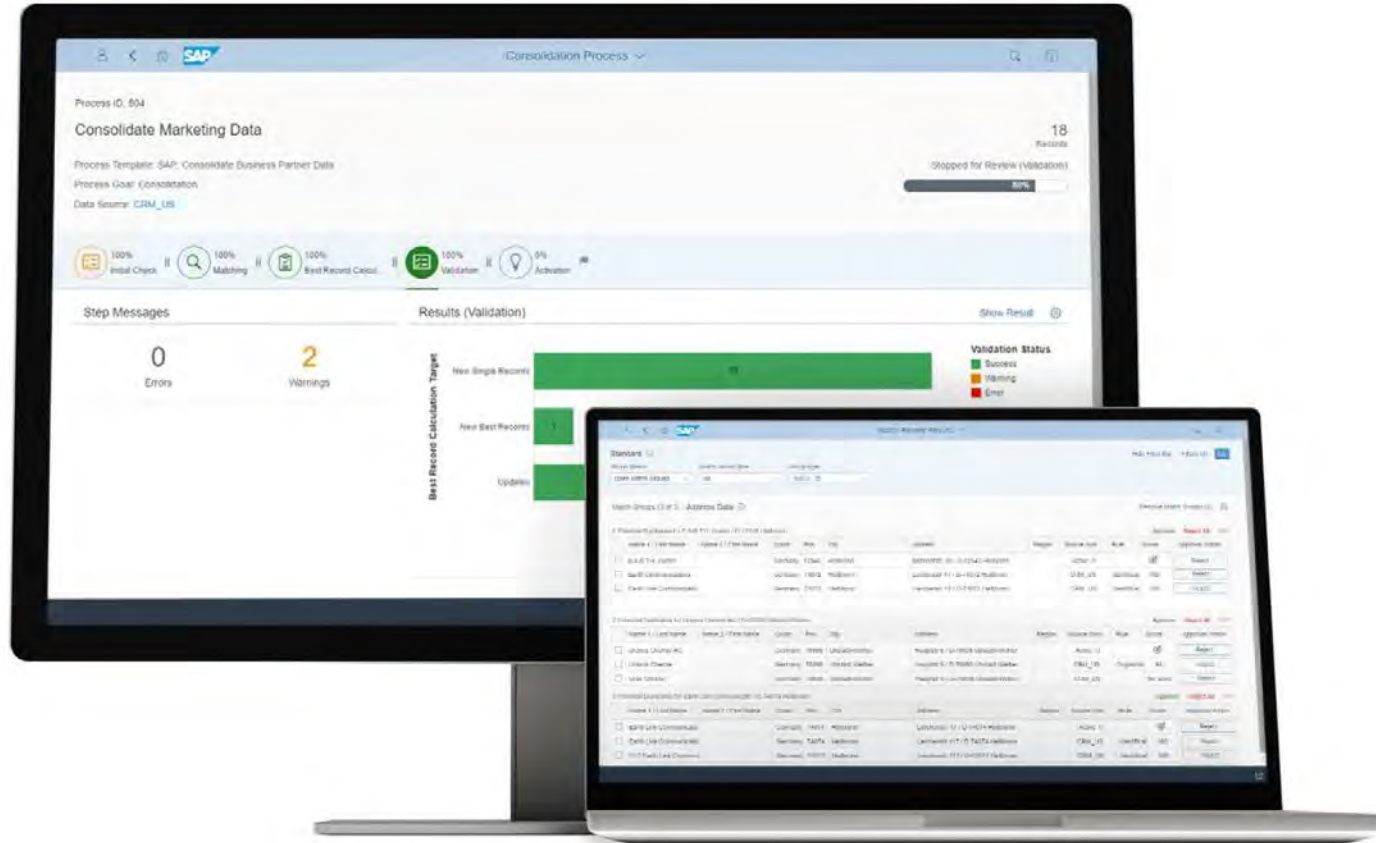
**Business process efficiency** to improve stability and quality of business process steps

**Reduced TCO** due to efficient data volume management

**Improved data exchangeability** between different solutions

**Lower risk** of breaching data privacy protection regulations

# SAP Master Data Governance



Consolidation

Central governance

Data quality & process  
Analytics



Process ID: 5254

## Load Runners Data

12936

Records

Process Template: SAP: Consolidate Business Partner Data

Stopped for Review (Matching)

Process Goal: Consolidation of Source Records

Data Source: REGISTERED\_RUNNERS

40%



## Step Messages

0

Errors

0

Warnings

## Match Group Information

| Status   | Groups | Records |
|----------|--------|---------|
| Open     | 3      | 8       |
| Approved | 3610   | 12828   |
| Rejected | 0      | 0       |

## Results (Matching)

[Match Review](#)

Open Match Groups

Approved Match Groups

## Group Type

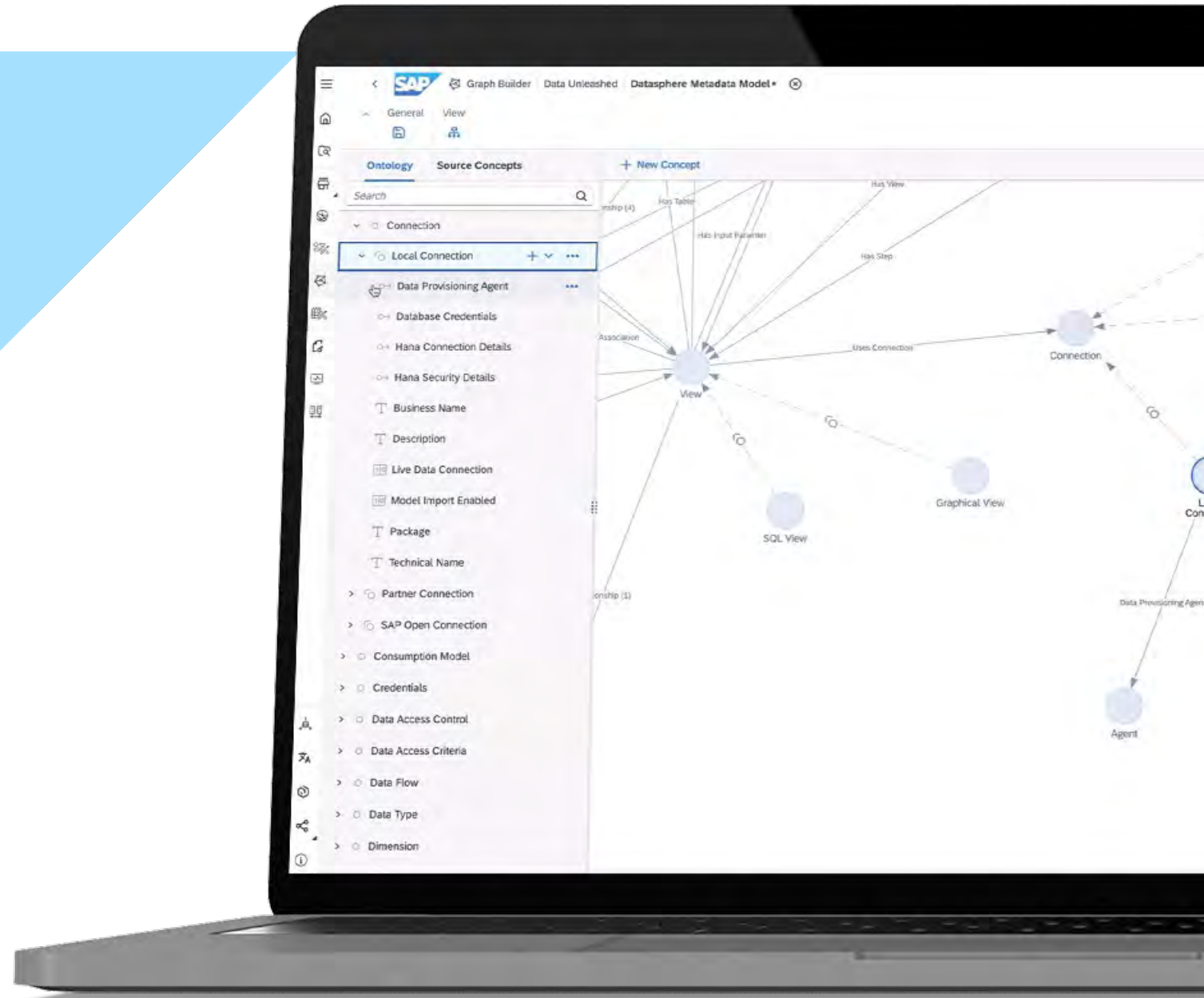
- With Active Records
- Source Records Only





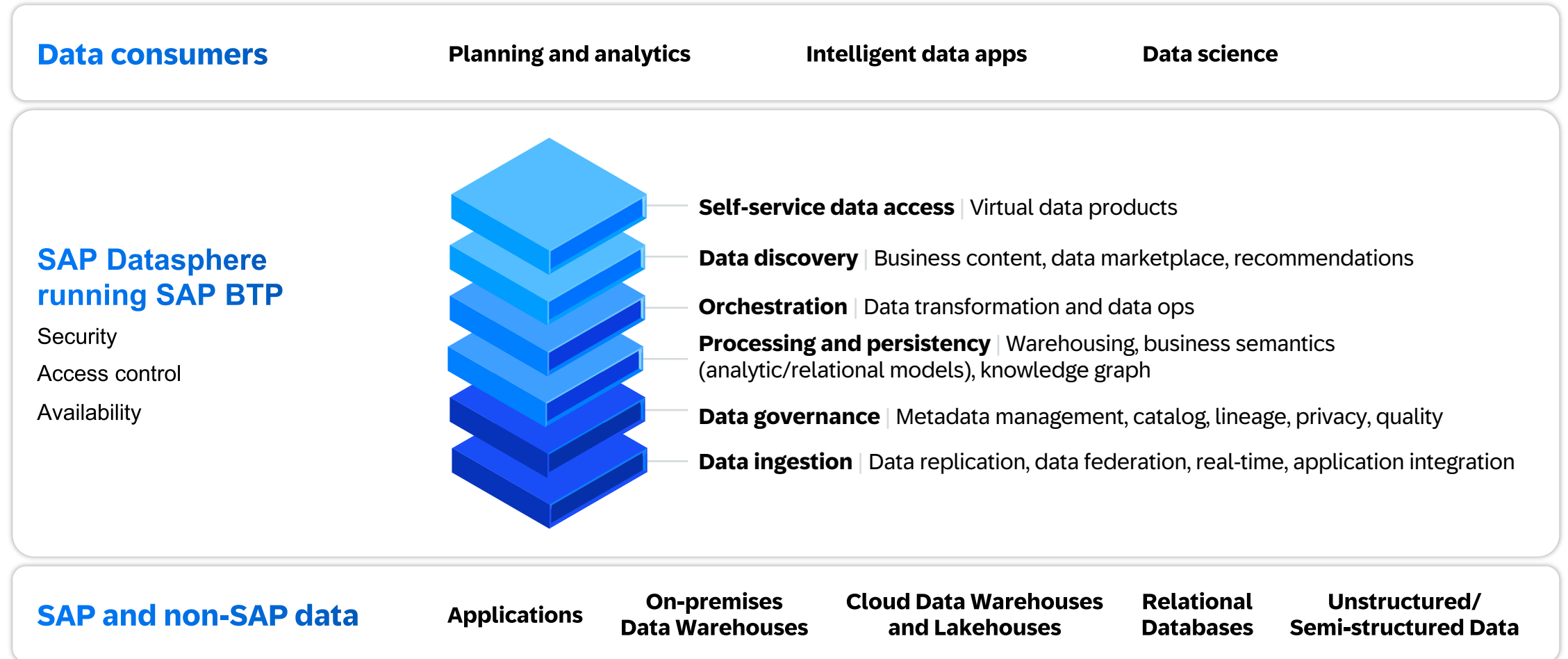
# SAP DATASPHERE

PUBLIC





# SAP Datasphere is the foundation for a business data fabric architecture





# Keep the **integrations** clean

## Main Aspects:

- ✿ **Integrations: standard API's**
- ✿ Loosely coupled integrations: **Event-Driven Design**
- ✿ Avoid **traditional API's** (RFC and IDoc)
- ✿ **SAP Application Interface Framework**

Keep the **extensions** clean

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# What makes integrations clean core compliant?

## What is “clean” integration?

### Main aspects

- **Base integrations on standard APIs** (OData and SOAP)
- **Aim for side-by-side extensibility with API integration or even SAP Cloud SDK** by utilizing the tight coupling with **SAP Integration Suite**
- Realize loosely coupled integrations in an **event-driven design** based on standard **events\***
- Avoid **traditional APIs** (RFC and IDoc) and their related **classical extension options**
- Ensure proper monitoring and error resolution capabilities using **SAP Application Interface Framework**

### How to achieve

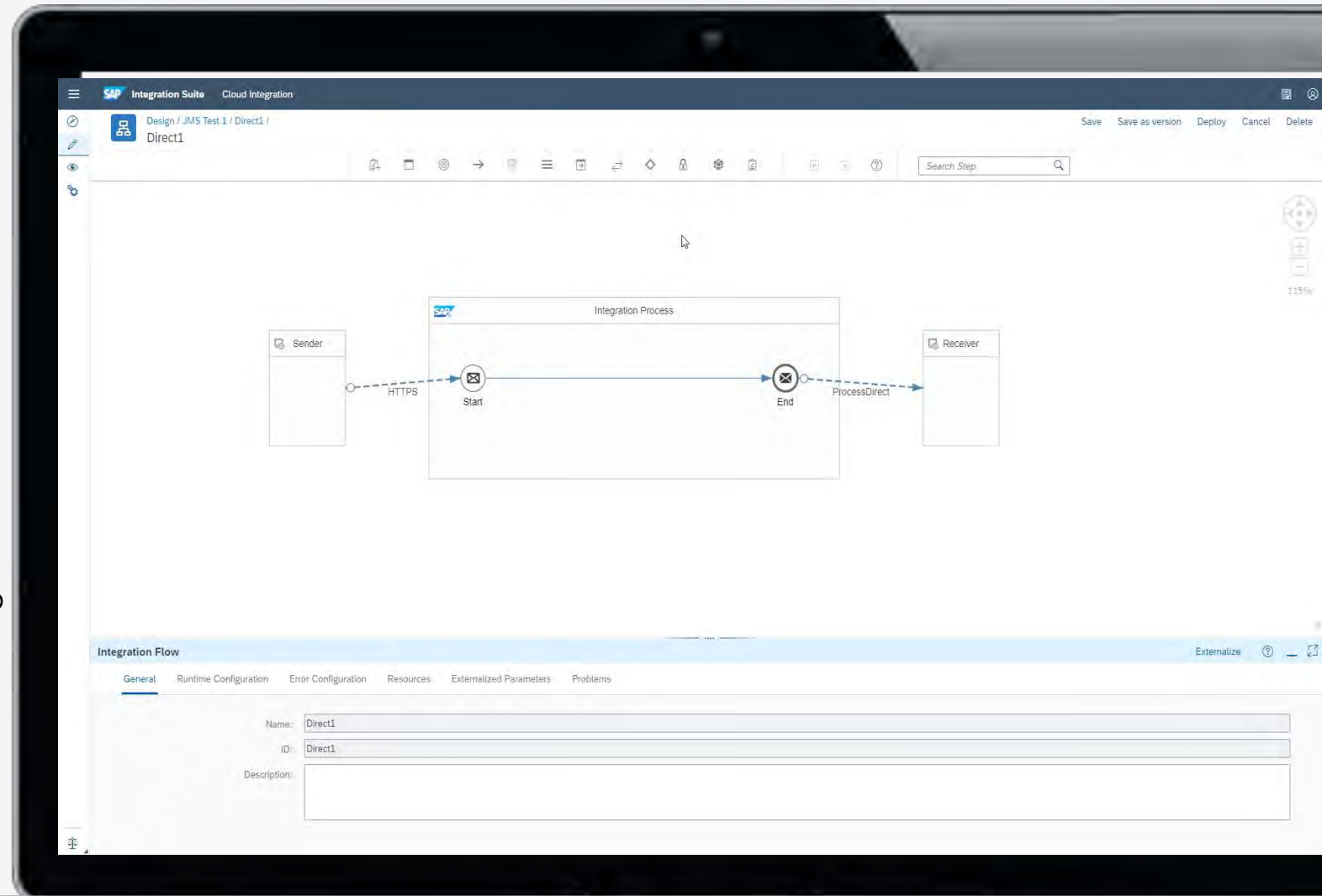
- Establish a clearly defined integration strategy with [SAP Integration Solution Advisory Methodology](#):
  - The central access point to discover integration artifacts like standard APIs, events, and integration flows is [SAP Business Accelerator Hub](#).
  - If mediation is required, use SAP Integration Suite to benefit from the tight integration with other SAP BTP capabilities.
- Define a one-time “get-clean” process:
  - Create an integration repository (included in [SAP Solution Manager](#)) to identify existing integrations and the technology or protocol on which they are based.
  - Establish “get clean” service inside the company to evaluate how utilized traditional APIs (RFC and IDoc) as well as their classical extensions could be converted into standard interfaces (fit to standard).
- Establish a continuous “keep clean” process (governance model):
  - Define central governance functionality for evaluation of new interface requirements or any interface adjustment based on defined SAP Integration Solution Advisory Methodology characteristics. The [Integration Assessment capability](#) could be used as an accelerator.
  - Apply a “keep clean” process for the most important and critical integrations.

**\*Event:** A data record expressing a significant change in state (for example, change of a business partner) and consisting of data representing the occurrence and context metadata. It is sent to an event provider such as the **SAP Event Mesh** capability, where consumers can subscribe to it. The business object data needs to be pulled by each consumer individually by using standard APIs.

Connect the  
disconnected

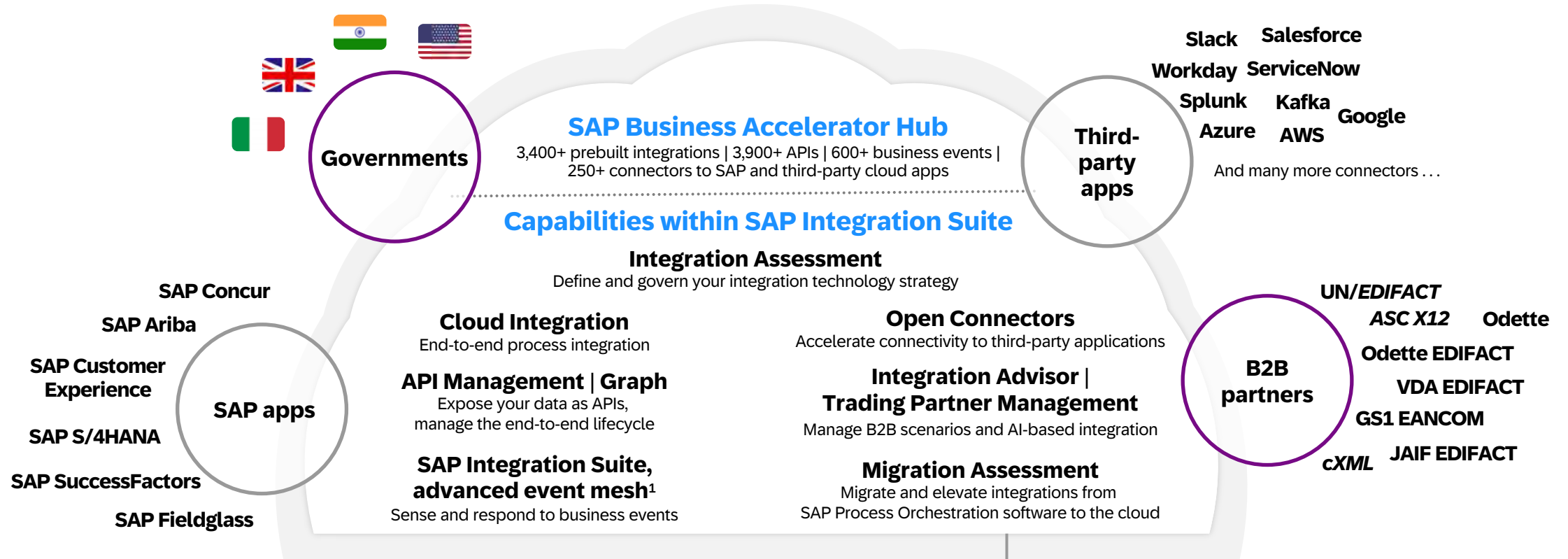
# SAP Integration Suite

Connect apps & processes with  
prebuilt tools & content across SAP  
& non-SAP.



# SAP Integration Suite

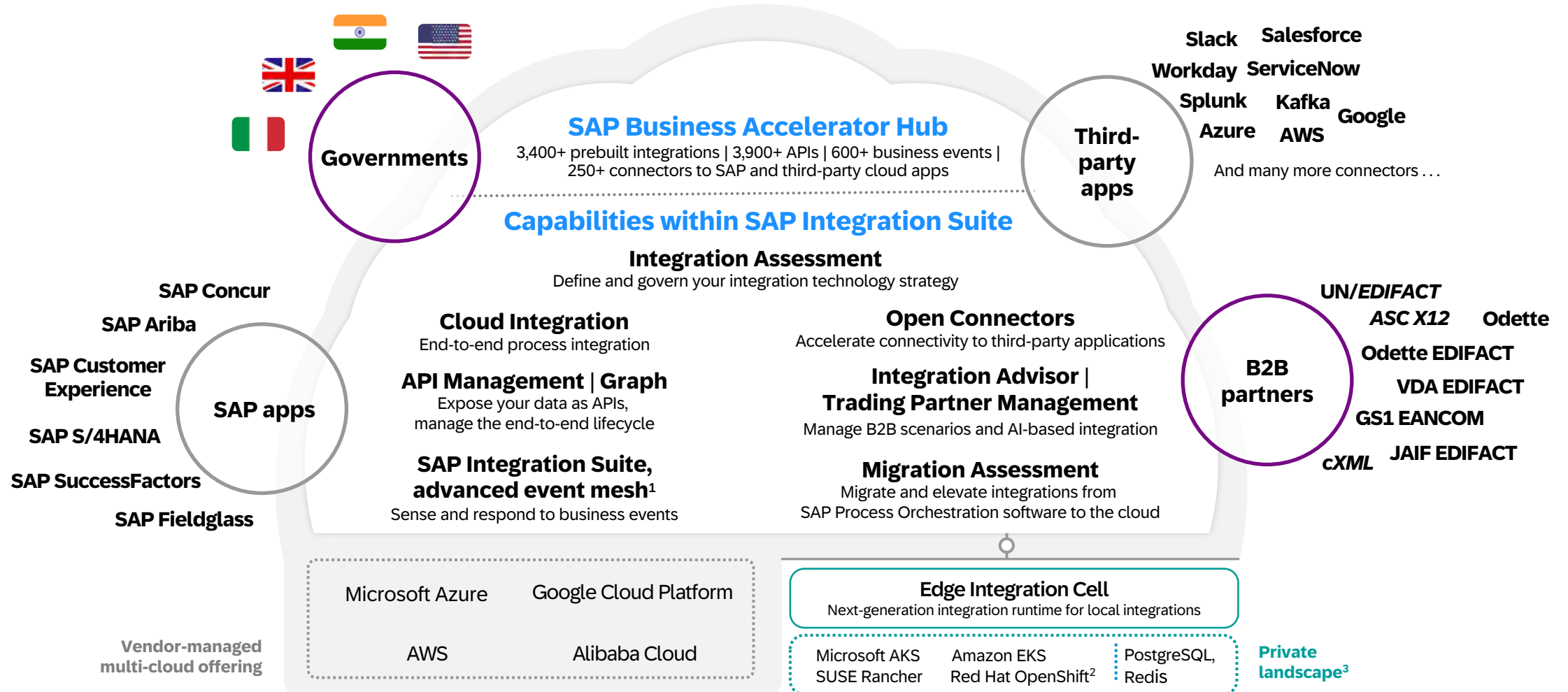
## SAP's hybrid integration platform for the Intelligent Enterprise



<sup>1</sup> Optional separate license | <sup>2</sup> Part of the road map, subject to change | <sup>3</sup> Private landscape corresponds to on premise or private cloud

# SAP Integration Suite

## SAP's hybrid integration platform for the Intelligent Enterprise



<sup>1</sup> Optional separate license | <sup>2</sup> Part of the road map, subject to change | <sup>3</sup> Private landscape corresponds to on premise or private cloud



SAP Business Accelerator Hub

Accelerate your integrations, extensions, and innovations

Search by product, package, or category

- Discover
- Products, Processes, and Partners.
- Explore
- Integrations, APIs, and Accelerators.
- Consume
- Visualize and consume integrations and workflows.

- Products
- Business Processes
- Categories
- Domains
- Industries
- Partners
- More
- Community

Choose a Product to Explore

See the various resources that each product has to offer

View All Products

SAP S/4HANA Cloud Public Edition

Ready-to-run cloud ERP that delivers the latest industry best practices and continuous innovations. You can also accelerate and grow your business with continuous innovations.

+7

SAP S/4HANA

A future-ready ERP system with built-in intelligent technologies, including AI, machine learning, and advanced analytics which transforms business processes with intelligent automation.

+5

SAP Customer Experience

Bring together customer data, machine learning technology, and microservices to power real-time customer engagements across sales, service, marketing, and commerce.

SAP Business Technology Platform

Accelerate business outcomes with integration, data to value, and extensibility.

+6

SAP SuccessFactors

A global, cloud-based human resource man-

SAP Ariba

A cloud-based innovative solution that al-

SAP Concur

Provides an integrated online and mobile

SAP Fieldglass

A cloud-based, open Vendor Management

FEEDBACK



# Keep competitiveness while reducing complexity

## Main aspects

- ✦ State-of-the-art process governance
- ✦ Application architecture
- ✦ SAP Best Practices packages
- ✦ Business process design
- ✦ Established process management

Keep the  
**extensions** clean

1

Extensibility

Keep the  
**data** clean

2

Data

Keep the  
**Integrations** clean

3

Integrations

Keep the  
**processes** clean

4

Processes

Keep the **operations**  
effective and efficient

5

Operations

# What makes **business processes** clean core compliant?

What is a “clean” process?

## Main aspects

- Establish **state-of-the-art business process governance**, including proper requirement management and transparent organizational structures
- Use recommended **SAP Reference Solution Architecture** to leverage **SAP solutions** such as SAP S/4HANA and others
- Implement respective **SAP Best Practices** where available and applicable
- **Tailor or enhance standard solutions from SAP** only where competitive advantages can be created
- Achieve **business process management excellence** by documenting processes and continuously monitoring process flow as well as process performance and efficiency

## How to achieve

- Governance model and organizational structures:
  - Proper requirement management includes a [solution standardization board](#) or a similar deciding structure, a suitable methodology (for example, SAP Application Extension Methodology – see slide “Clean extensions”), and proper documentation and tooling support such as the [SAP Cloud ALM solution](#) or the [Focus Build solution](#) for [SAP Solution Manager](#).
  - A [business process management](#) structure is established and transparent across relevant organizations, including process owners and related contact persons.
- The solution architecture can be based on the [SAP Enterprise Architecture Framework](#) methodology. Supporting resources can be leveraged using the [SAP Transformation Navigator tool](#), [SAP Signavio Process Explorer solution](#), [SAP Signavio Process Manager solution](#), [SAP Signavio Process Collaboration Hub](#), and the [partner solution LeanIX](#).
- SAP Best Practices can, in general, be compared to SAP Signavio Process Explorer, the [SAP Signavio Process Navigator solution](#), SAP Signavio Process Manager, or SAP Signavio Process Collaboration Hub. Find more general guidance in the [Administration Guide to Implementation of SAP S/4HANA with SAP Best Practices](#).
- For individual implementation projects, the [SAP Activate](#) innovation adoption process and the related [road map viewer](#) can be leveraged. Further preconfiguration is provided through the [enterprise management layer for SAP S/4HANA](#). Industry-specific best practices can be requested through SAP standard content activation service.
- The solution design is focused on the following principles:
  - For nondifferentiating processes, SAP standard is strictly applied; for differentiating processes, SAP standard solutions are tailored to business needs and enhanced where needed.
  - A focus on mandatory and key process variants helps avoid unnecessary customizing and process complexity.
  - For required extensions the recommendations for “clean extensions” are followed.
- Business process management excellence covers the following three aspects:
  - Customer-specific process design is documented (SAP Signavio Process Manager and SAP Signavio Process Collaboration Hub).
  - As-is process execution can be measured ([SAP Signavio Process Insights](#), [SAP Signavio Process Intelligence](#), and using the [plug and gain approach](#)).
  - Business processes are efficiently executed based on measurable process performance indicators (SAP Signavio Process Insights and SAP Signavio Process Intelligence).



# Enterprise Architecture Blog Posts

Need a little more room to share your thoughts with the community? Post a blog in the SAP Enterprise Architecture group to explain the more complex topics.

All community ▾

What are you looking for today?

🔍



## SAP Enterprise Architecture Framework



 **Rene\_de\_Daniel**

Product and Topic Expert



11-08-2022 4:49 PM



33 Kudos

This article will provide further insights into the SAP Enterprise Architecture Framework which was presented by Andreas Poth at the [SAP Enterprise Architecture Global Summit](#) (March '22) and the [TOGAF® Standard 1010th Edition – Launch event](#) (May '22).

**The need for a common and aligned framework**

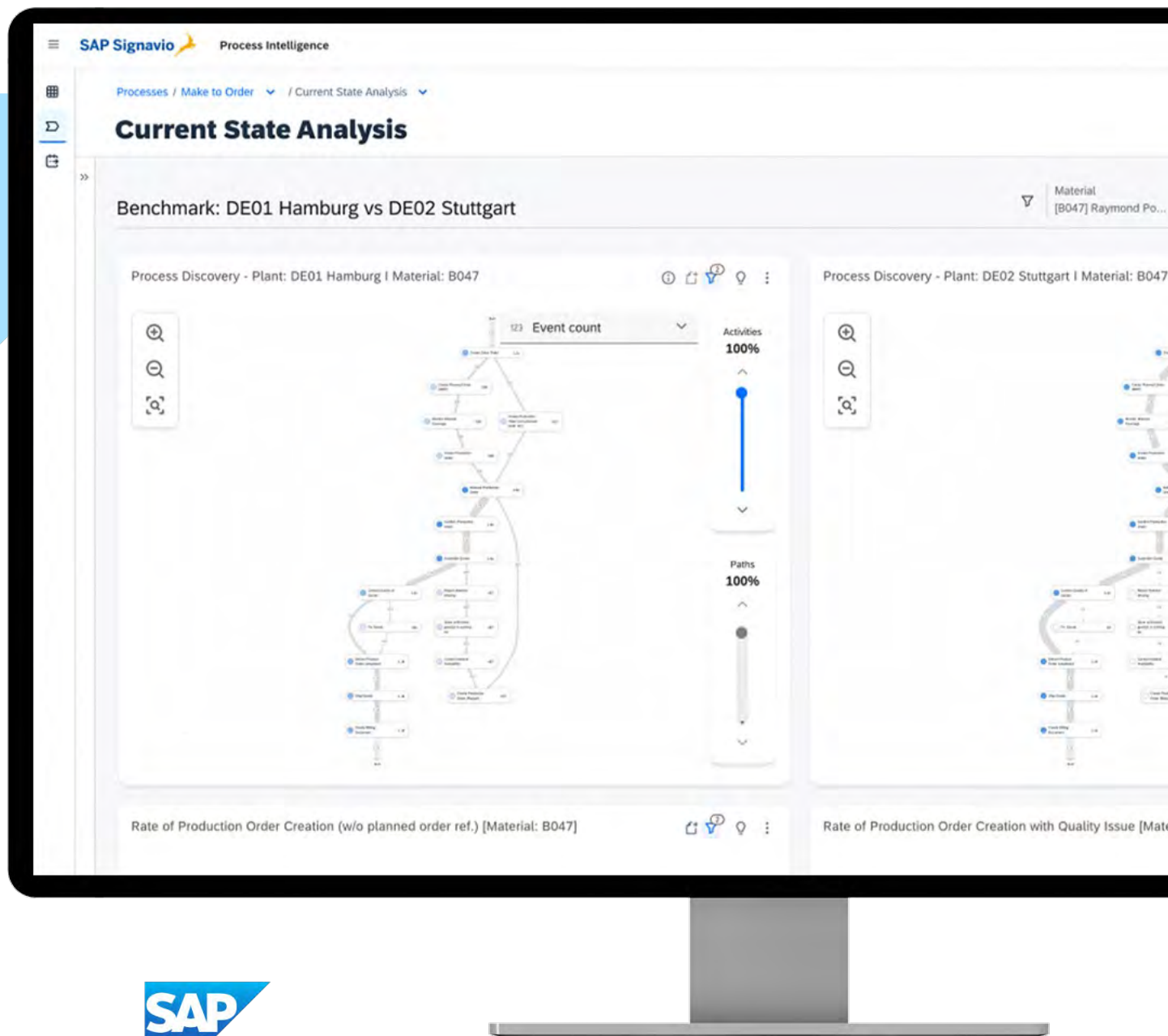
## Labels In This Area

- Application Architecture 23
- Business Architecture 33    Data Architecture 19
- Emerging Trends 20
- Enterprise Architecture 55    Frameworks 22
- Hybrid and Multi Cloud 3    Innovation 14
- Integration Architecture 18    Portuguese 1
- Roadmaps 12    Skills and Learning 29
- Solution Architecture 22    Sustainability 3
- Technology Architecture 24    Tools 14

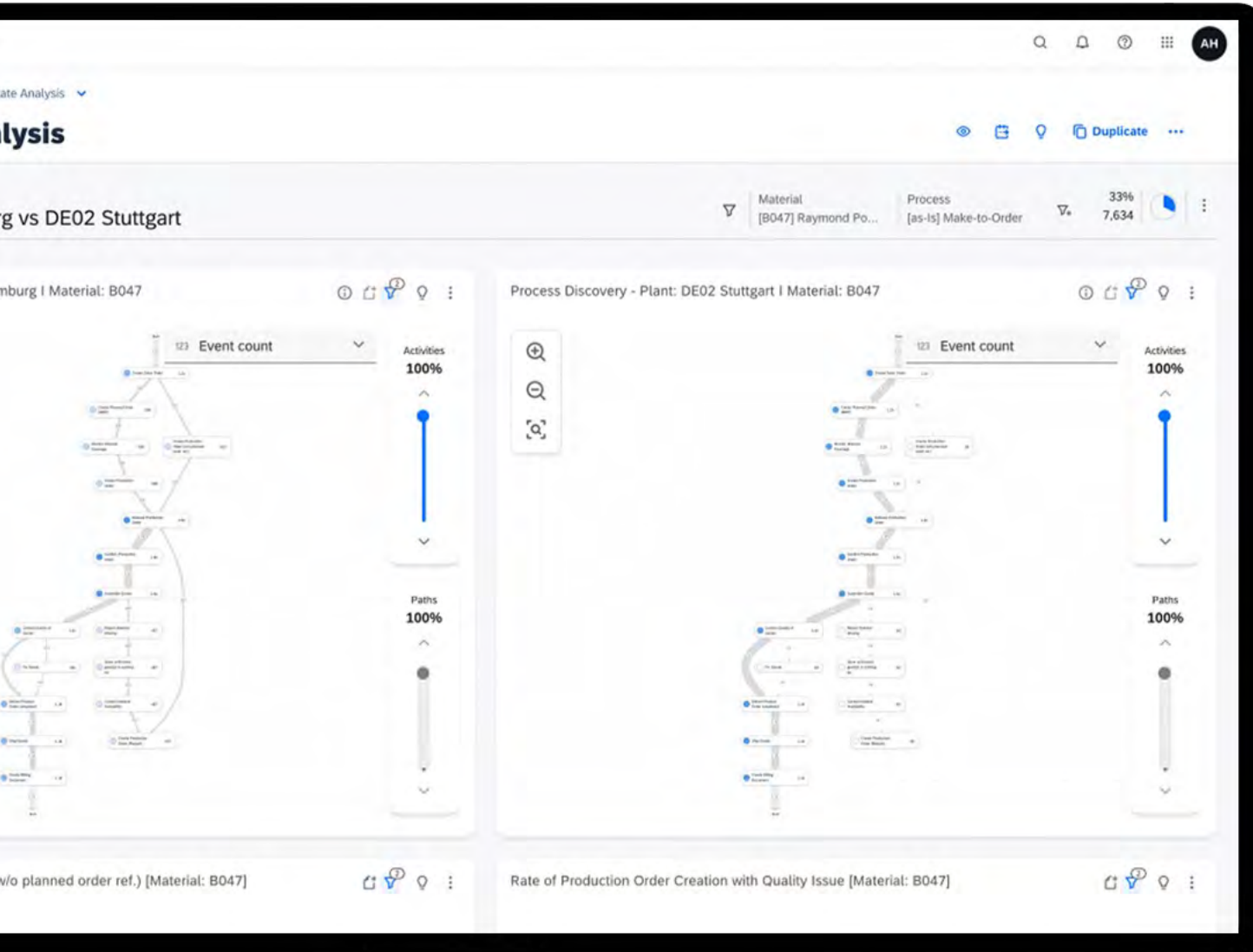
# SAP Signavio

## Process Insights & Process Intelligence

PUBLIC



# SAP Signavio



Business Process  
Management

Process Mining and  
recommendations

Process best practices





# Keep operations **effective and efficient**

## Main aspects

- ⚙ **End-to-end concept for operations.**
- ⚙ **Release management**
- ⚙ Housekeeping activities that are in line with SAP Best Practices

Keep the **extensions** clean

1

Extensibility

Keep the **data** clean

2

Data

Keep the **Integrations** clean

3

Integrations

Keep the **processes** clean

4

Processes

Keep the **operations** effective and efficient

5

Operations

# What makes operations clean core compliant?

## What are “clean” operations?

### Main aspects

- The paradigm of keeping the core clean is integrated into the **end-to-end concept for operations**.
- **Release management** is an established foundation for a clean core; the latest release should always be targeted.
- **Housekeeping** activities that are in line with SAP Best Practices are pursued, and the distribution of roles and responsibilities that are agreed with SAP are followed.
- It is agreed that SAP performs maintenance for technology within the preapproved **CMPs**.

### How to achieve

- Establish keeping the core clean as an integral part of the end-to-end operations concept:
  - Consider “keep clean” as an IT service to add business value and establish IT as a service provider to own the end-to-end view and end-to-end processes.
  - Integrate monitoring and alerting of a “keep clean” process into the overall concept for operations to have an integrated view on KPIs for affected areas that define a clean core (integration, extensibility, processes, data).
  - Establish procedures for event management and escalations that are in line with the established governance models for integration, extensibility, processes, and data. Consider using SAP’s operations platforms to achieve this – [SAP Cloud ALM](#) or [SAP Solution Manager](#).
- Release management is based on two core principles:
  - CMPs create preapproved monthly maintenance windows that are agreed by all involved parties and adequately documented.
  - Only in exceptional rare cases, it is requested to skip maintenance windows (“opt out”).
- Regular housekeeping is established and supports keeping the core clean, for example:
  - Background job management (including approval, documentation, monitoring, and improvement) is implemented to contribute to an efficient utilization of infrastructure.
  - The usage of file interfaces for importing or exporting data from or into the system is avoided to strengthen security, maintainability, and consistency.
  - End-user authorizations are reviewed and adapted on a regular basis. Unneeded authorizations are unassigned from users and discontinued.

# SAP Cloud Application Lifecycle Management

# HITACHI

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Leveraging SAP Business Technology Platform to accelerate move to the cloud and keeping the core clean

[Watch the video](#)

[Read the Story](#)



Reduced custom ERP add-ons

## 94% fewer

SAP ERP add-ons  
decreased from over 9,000  
to 520

Accelerated S/4HANA Cloud upgrade cycles

## 1 month

Time spent on upgrading  
ERP versions cut from one  
and a half years to approx. one month

# MYTHS & FACTS

*"BTP is the only means of making and keeping the core clean"*

*"Clean Core is only about Custom Code and Extensibility."*

*"Fit-to-standard is the only means of making and keeping the core clean"*

*"Clean Core is only about TCO"*

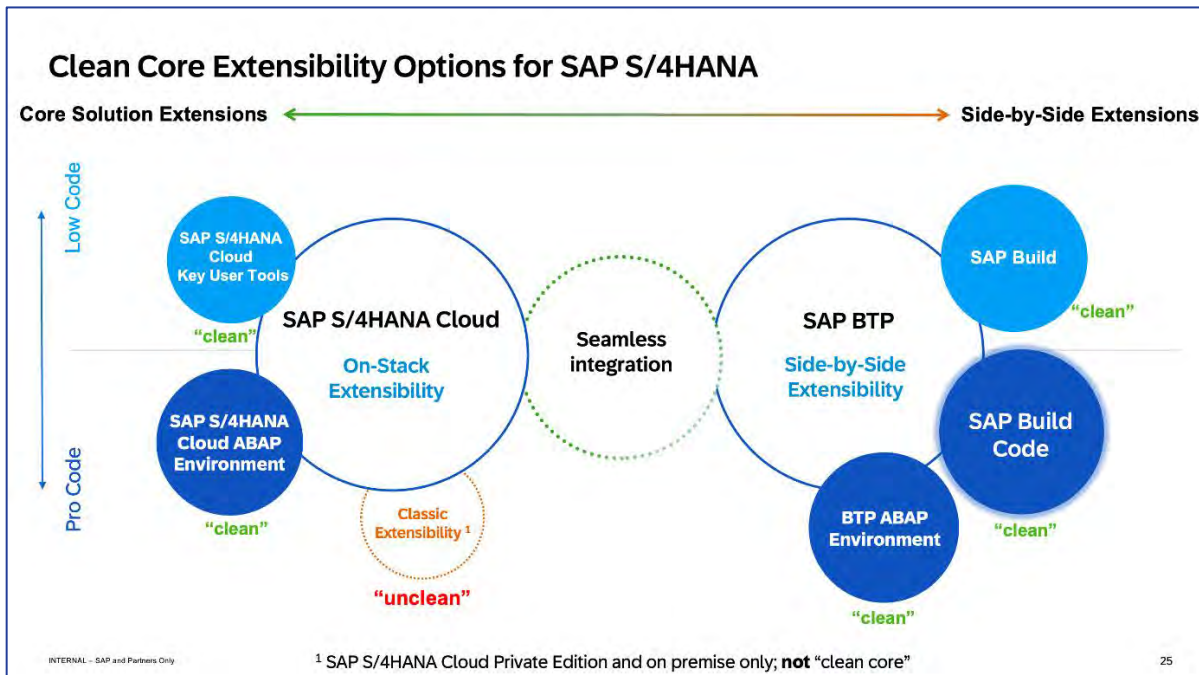


# MYTHS

&

# FACTS

*"BTP is the only means of making and keeping the core clean"*



A Clean Core can be achieved by:

1. On-stack (key-user & developer extensibility)
2. Side-by-side extensibility (BTP)
3. Fitting processes to standard or
4. A mix of all of the above

# MYTHS

&

# FACTS

*"Clean Core is only about Custom Code and Extensibility."*



Clean Core affects:

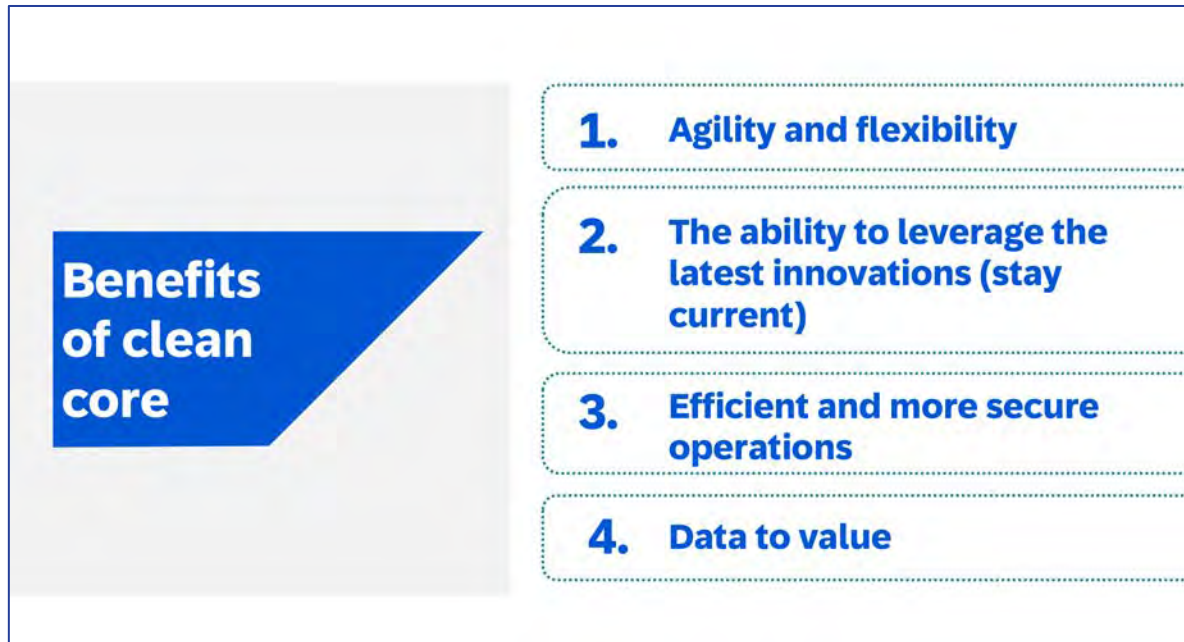
- Custom Code i.e. Extensibility as well as
- Data,
- Processes,
- Integration and
- Operations

# MYTHS

&

# FACTS

*"Clean Core is only about TCO"*



Clean Core not only contributes to a TCO reduction:

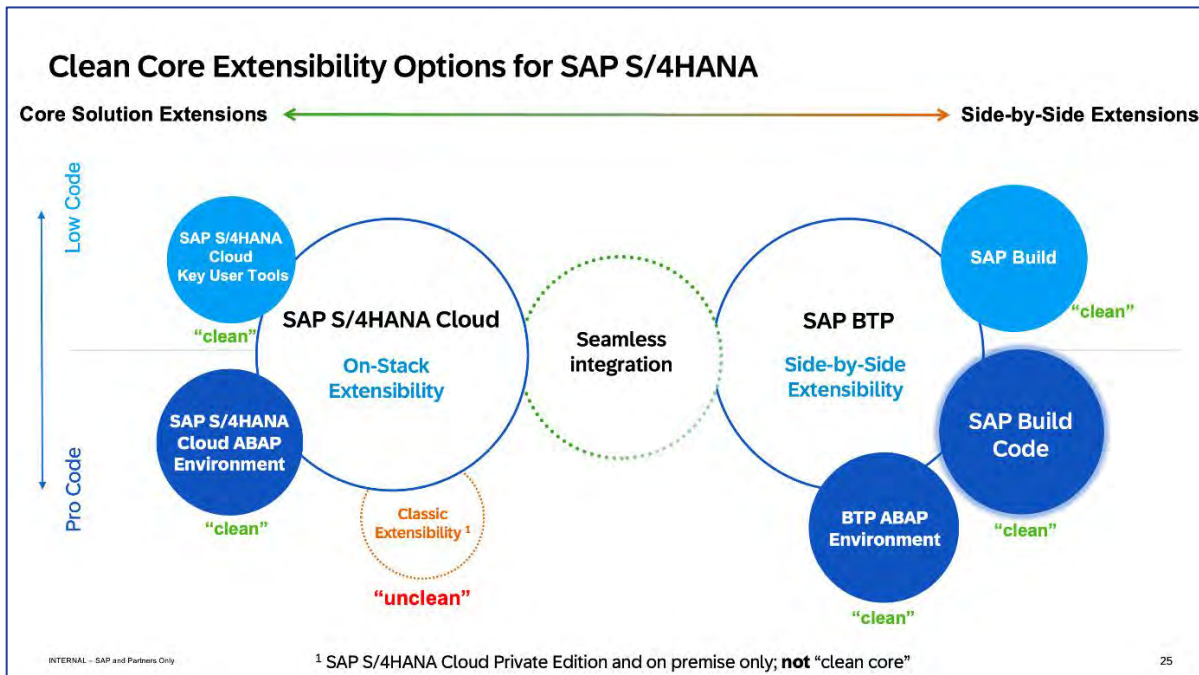
- It also enables the consumption of the latest innovations
- A mindset and paradigm shift from OnPrem to Cloud
- Provides you with more flexibility in your business

# MYTHS

&

# FACTS

*"Fit-to-standard is the only means of making and keeping the core clean"*



- Although a fit-to-standard system is probably mostly clean
- the usage of "on-stack" and "Side-by-side" extensibility options also make and keep the core clean.

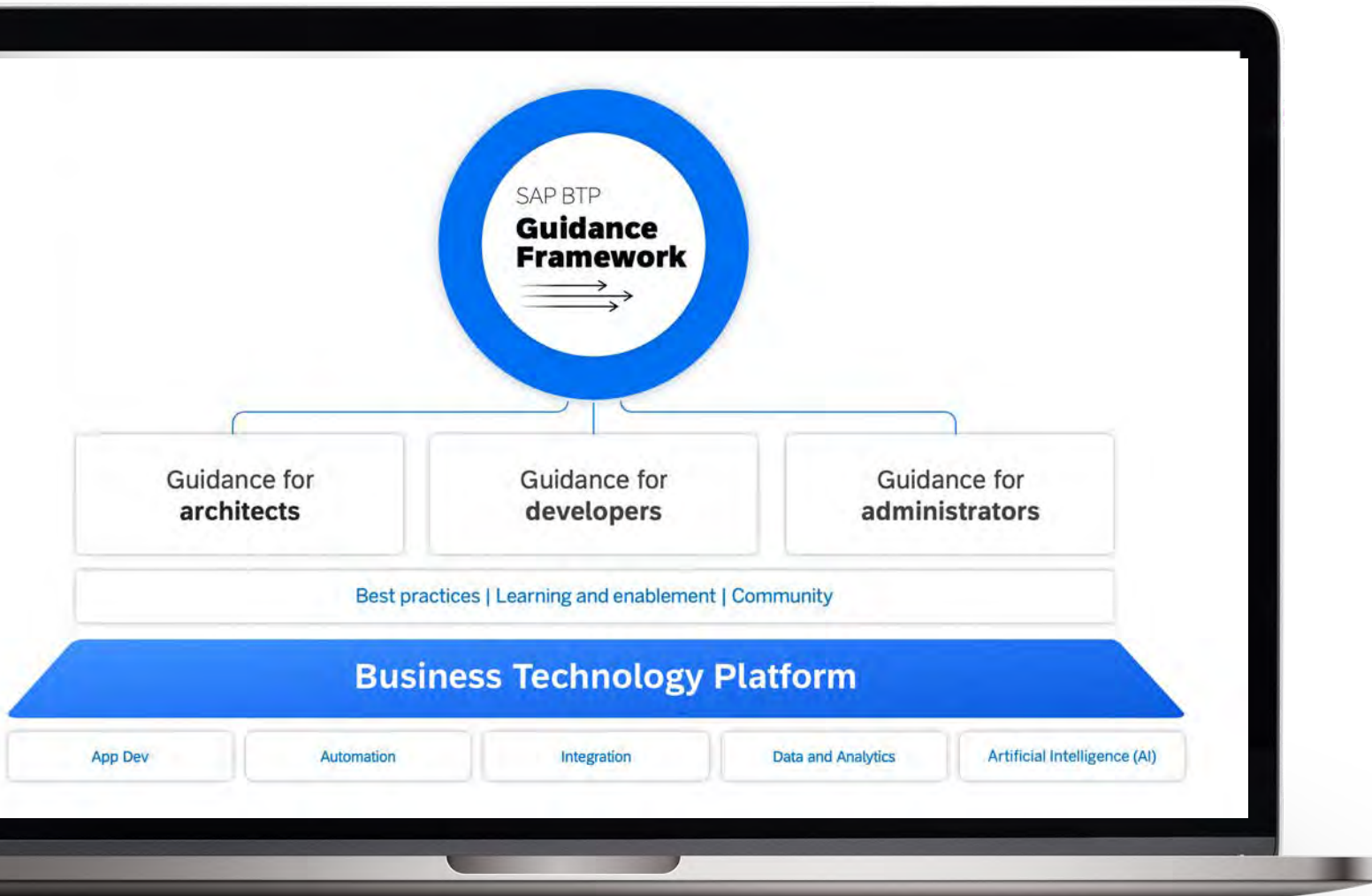


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# How to start from here?



# SAP BTP Guidance Framework



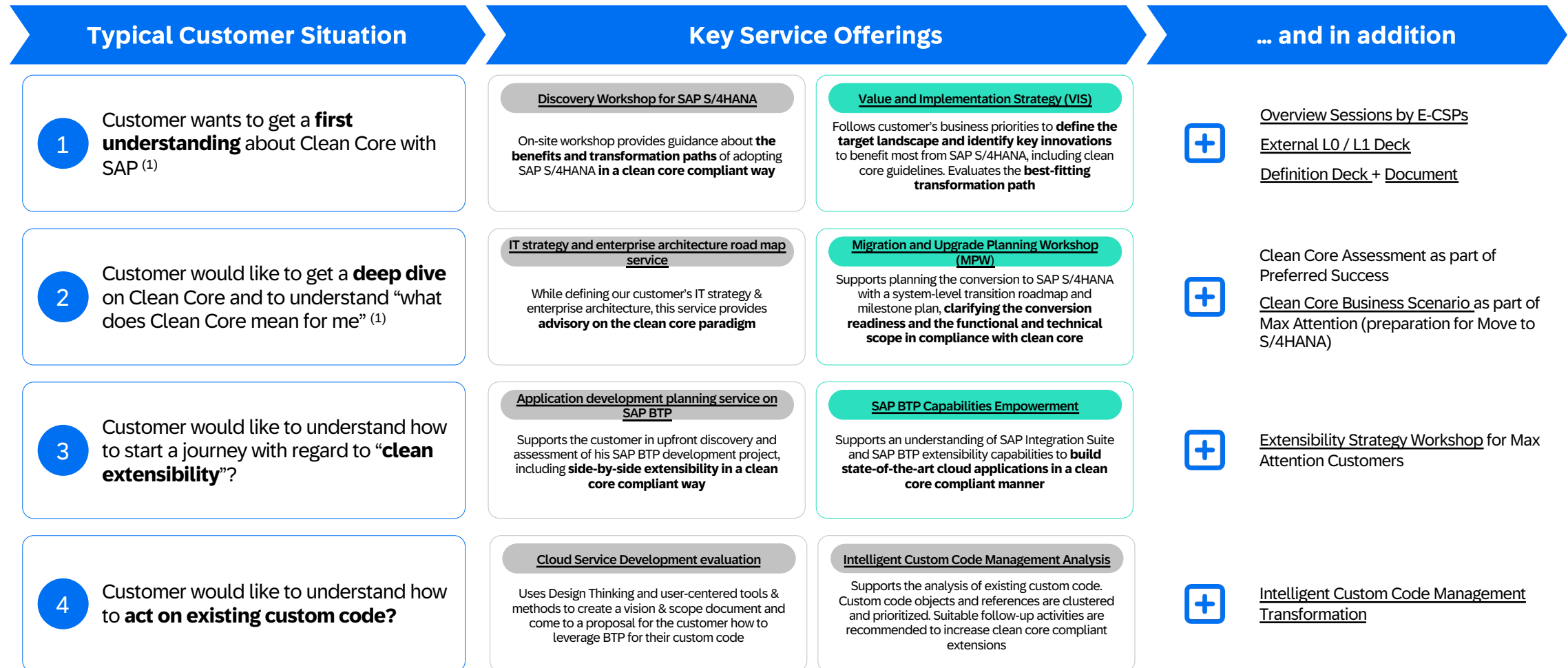
## SAP BTP Guidance Framework



The SAP BTP Guidance Framework provides a central access point for architects, developers and administrators to build and run enterprise-grade solutions on SAP BTP.

# Key SAP Services Offerings for Clean Core

## Typical customer situations (1/2)



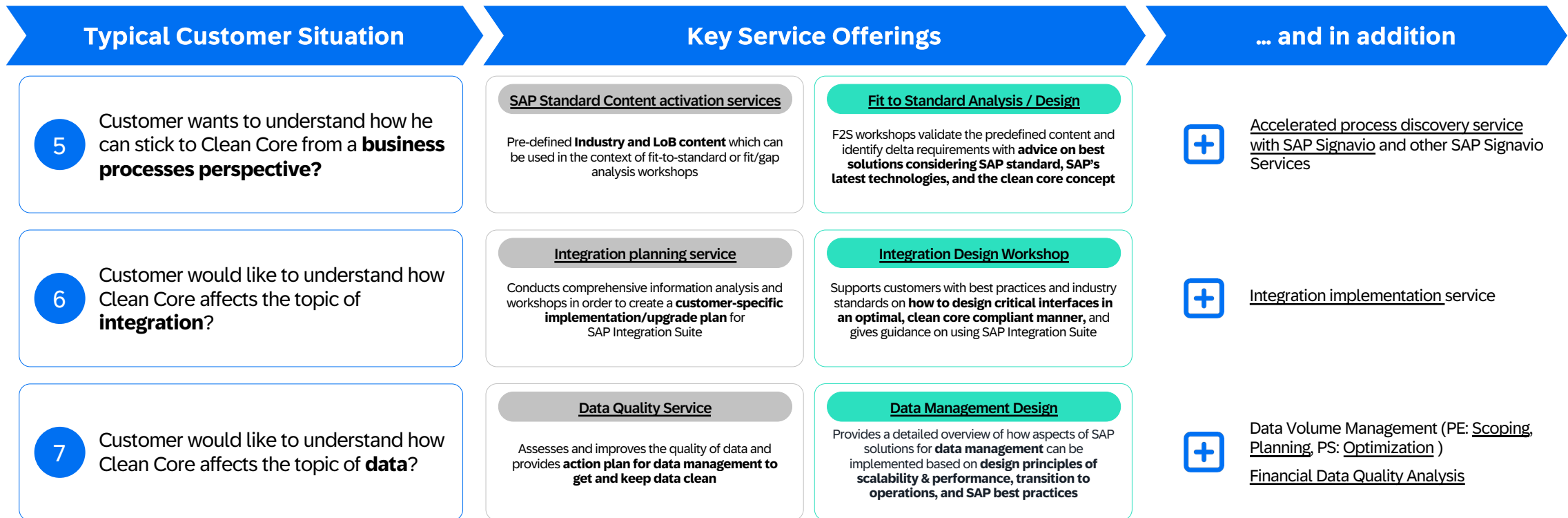
<sup>(1)</sup> for customers planning a journey to SAP S/4HANA. If the customer is already on SAP S/4HANA, start from situations 3-7.

- Services available Professional Services customers
- Services available only for Premium Engagement customers



# Key SAP Services Offerings for Clean Core

## Typical customer situations (2/2)

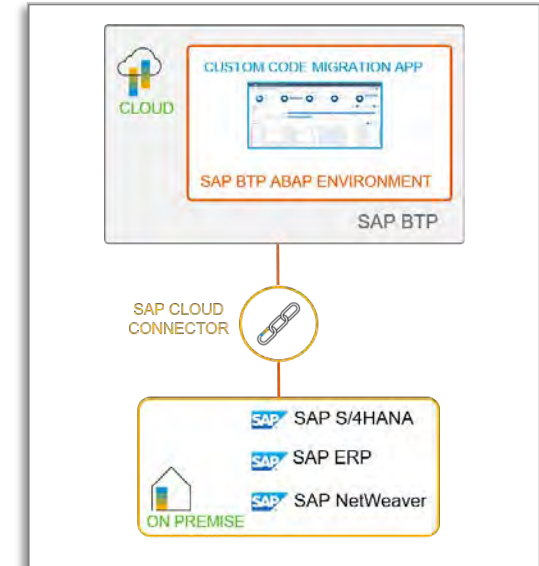


- Services available Professional Services customers
- Services available only for Premium Engagement customers

# Custom code analysis with Custom Code Migration app

## SOLUTION OVERVIEW

- Available in an SAP S/4HANA >=1809 system or in SAP BTP ABAP Environment
- Technically based on [remote ABAP test cockpit](#)
- Analysis of custom code for SAP S/4HANA, SAP BTP, ABAP Cloud, classic ATC use cases
- Simplified custom code remediation
- Obsolete custom code removal based on usage data
- Identification of custom code candidates for back-to-standard and redesign
- Evaluation of custom code for semi-automated adaptation (quick fixes in ABAP development tools for Eclipse)



## HIGH LEVEL FEATURES

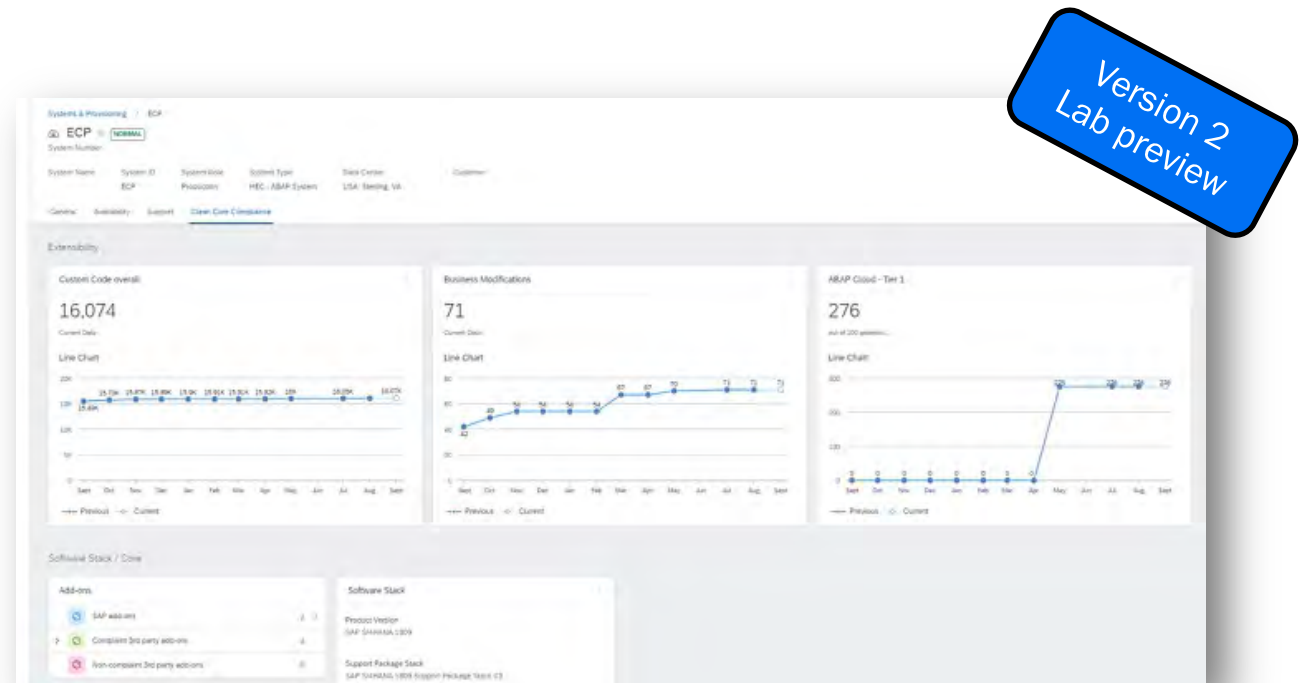
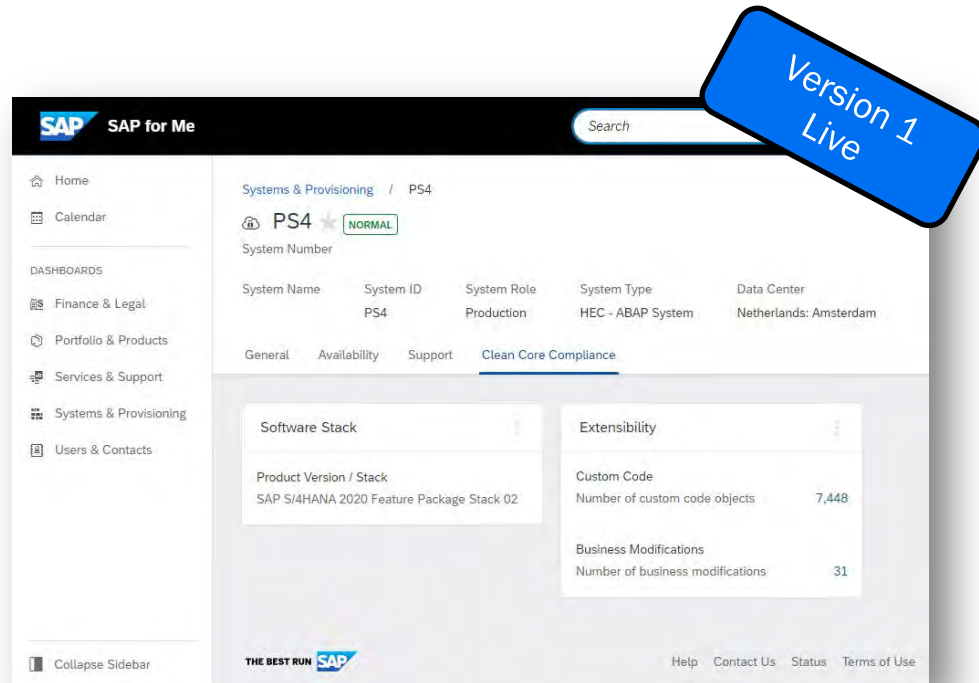
### SCOPING



### ANALYSIS



# Self Service – Transparency - Clean Core Dashboard



- Available for productive SAP S/4HANA Cloud, private edition systems in SAP for Me (Systems & Provisioning area)
- Gives a technical overview on the “clean core compliance” of your SAP S/4HANA Cloud, private edition systems
- Includes link to guides, services and tools (SAP and partner) that help to get closer to a clean core.